Women's Liberation and the Demographic Transition

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Introduction

- How does women's empowerment affect the household's fertility choices?
- 'Coverture' was a legal doctrine in common law that limited legal/economic status of married women.
- Its repeal was a dramatic change of power within households.
- "*If the principle of* séparation des biens *were to be made a rule of law instead of an exception, our whole social relations would be changed. Old-fashioned people like himself were not ashamed to declare that it was written in nature and in Scripture that the husband was and ought to be lord of his household, the regulator of its concerns, and the protector of its inmates, which, if this Bill passed, he would no longer be.*" MP Beresford Hope, 1870.
- Use cross-state variation in the timing of coverture abolition in the US.
  - Full US Census, 1850-1920, comparing county-border pairs, using event study approach.
  - Full US Census, 1900-1910, comparing couples married before/after rights.

Coverture & Economic Rights

- Property Laws:
  - "Moveable", or "personal", assets, such as money, stocks, bonds, became the husbands' after marriage.
  - "Real" assets, such as land & structures, remained in the wife's name, but under the husbands' control.
- Earning Laws:
  - Wife's income belongs to husband.
  - Married women not allowed to contract.
- Transfers of property not undone with rights.
  - Compare couples married before/after.

Coverture in AU

Outline of Talk/Preview of Results

- Legal history: issues in timing of rights, “both” rights needed.
- Outline empirical approaches.
- Document that:
  - No impact of rights on marriage market.
  - Rights led to about a 5-7% decline in fertility.
    - That’s 15–20% of decline in this time period
- Mechanism: Bargaining power fits data/literature/historical anecdotes.
  - Married before/after
  - Maternal mortality risk
  - People with wealth, NYTimes coverage

Outline of Talk/Preview of Results 2

- Other stories don't work:
  - $\uparrow$ cost of women's time $\rightarrow \downarrow$ fertility? No rise in LFP.
  - G.E. effects? Hazan et al. (2019) document economic growth w/rights. $\downarrow$ fertility and $\uparrow$ education? Not consistent with only/mostly affecting those married after rights.

Literature

- Women's Economic Rights.
  - Causes: Geddes & Lueck (2002), Doepke & Tertilt (2009), Fernandez (2014)
  - Consequences: Kahn (1996), Roberts (2007), Geddes et al. (2012), Hazan et al. (2019).
- Laws and bargaining power in marriage.
  - Wolfers (2006), Stevenson and Wolfers (2006), Voena (2015).
- Bargaining power and fertility/education.
  - Thomas (1993), Lundberg et al. (1997), Attanasio & Lechene (2002), Rasul (2008), Qian (2008), Bobonis (2009), Albanesi & Olivetti (2014, 2016), Ashraf et al. (2014, 2020), Doepke & Tertilit (2018, 2019), Doepke & Kindermann (2019).
- Demographic transition: theory and application to the U.S.
  - Galor and Weil (2000), Galor and Moav (2002), Greenwood and Seshadri (2002), Doepke (2005), Bleakley and Lange (2009), Beach and Hanlon (2023).

Debt Statutes

- Panic of 1837.
- States: help debtors & women/children of bankrupt husband.
- Protect wife's real & moveable/personal assets from husband creditors.
- Wife had moveable if husband had not "reduced" it to possession. Definition by state. Legal issues.
- Not relevant for us. Koudijs & Salisbury have a great series of papers.

Property Rights

- Property rights. Complicated history. Example: New York.
- *Dickerman v. Abrams*, 1854 NY Supreme Court:
  - 1st attempt in 1848: "The disposition of her personal property and of the rents, issues, profits of her real estate had been taken from her husband, and lodged nowhere."
  - 2nd attempt in 1849: Wife could "…convey and devise real and personal property …".
  - However, "The words 'convey and devise' are technical terms relating to the disposition of interests in real property. It could not be technically or legally correct to speak of *conveying* personal property …or of *devising* it …".
- 1860: Gave women rights over personal property (and earnings).
- NJ, VA, WI, WV, …copied NY's laws, often verbatim.
- Generally: passed laws, courts interpreted conservatively, repeat.

Community

Confusion: Property rights, but not earnings rights.

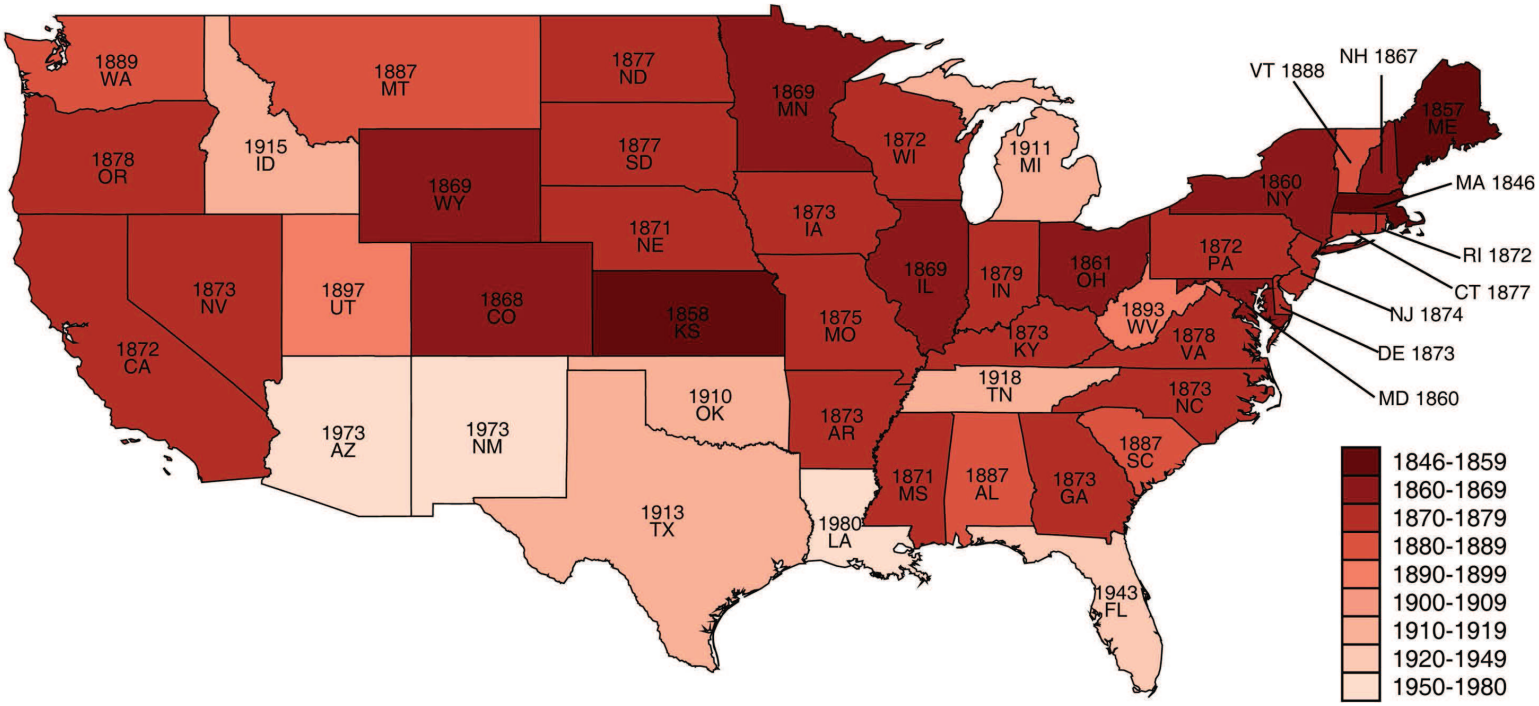
- *Glover v. Alcott* 11 Mich. 470 (1863).
  - Deborah Alcott owned land & mill.
  - Her husband declared bankruptcy.
  - His creditors demanded her profits (labor income).
  - Courts agreed - otherwise, set up pass-through business. Dissent pointed out that this implicitly meant she could not run a mill on her land.
- *Apple & Co. v. Ganong* 47 Miss. 189 (1872).
  - Louisa Ganong's husband declared bankruptcy.
  - Louisa had land purchased with gift from mother and earnings from civil war job.
  - Creditors got share of land in proportion to earnings from job.

Confusion: Earnings rights, but not property rights.

- Wife earned money, then it gets mixed into general pool.
- Courts were very hesitant to recognize women rights over earnings b/c the husband had right to wife's work at home (Stanley 1988).

  $\Rightarrow$ Need both rights.

Timing of Women's Rights by State: (Geddes & Lueck 2002)



Suffrage

Sample

Event Study Approach:

- Full count US census, 1850-1920.
  - LFP variables only 1860-1920.

Married Before/After Rights Approach:

- Duration of marriage ("married-after") only in 1900-1910.

Sample Selection (both empirical strategies):

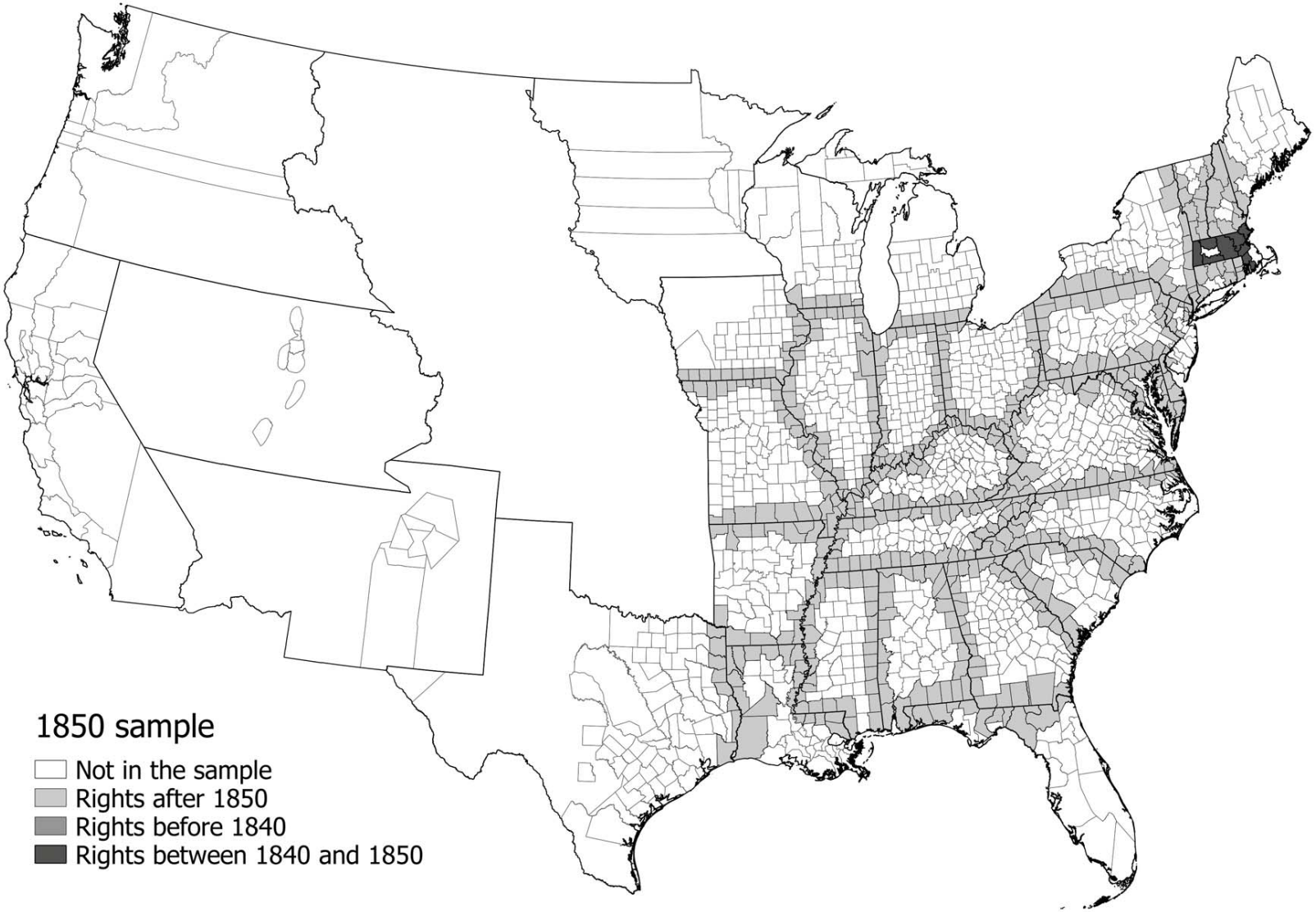
- All U.S.-born, married White, non-Hispanic (WNH) women aged 20 to 39, with husbands up to age 49
- "rights" are assigned by state of birth.
- Robustness: All women in the above group who reside in their state of birth

Event Study: 1850-1920

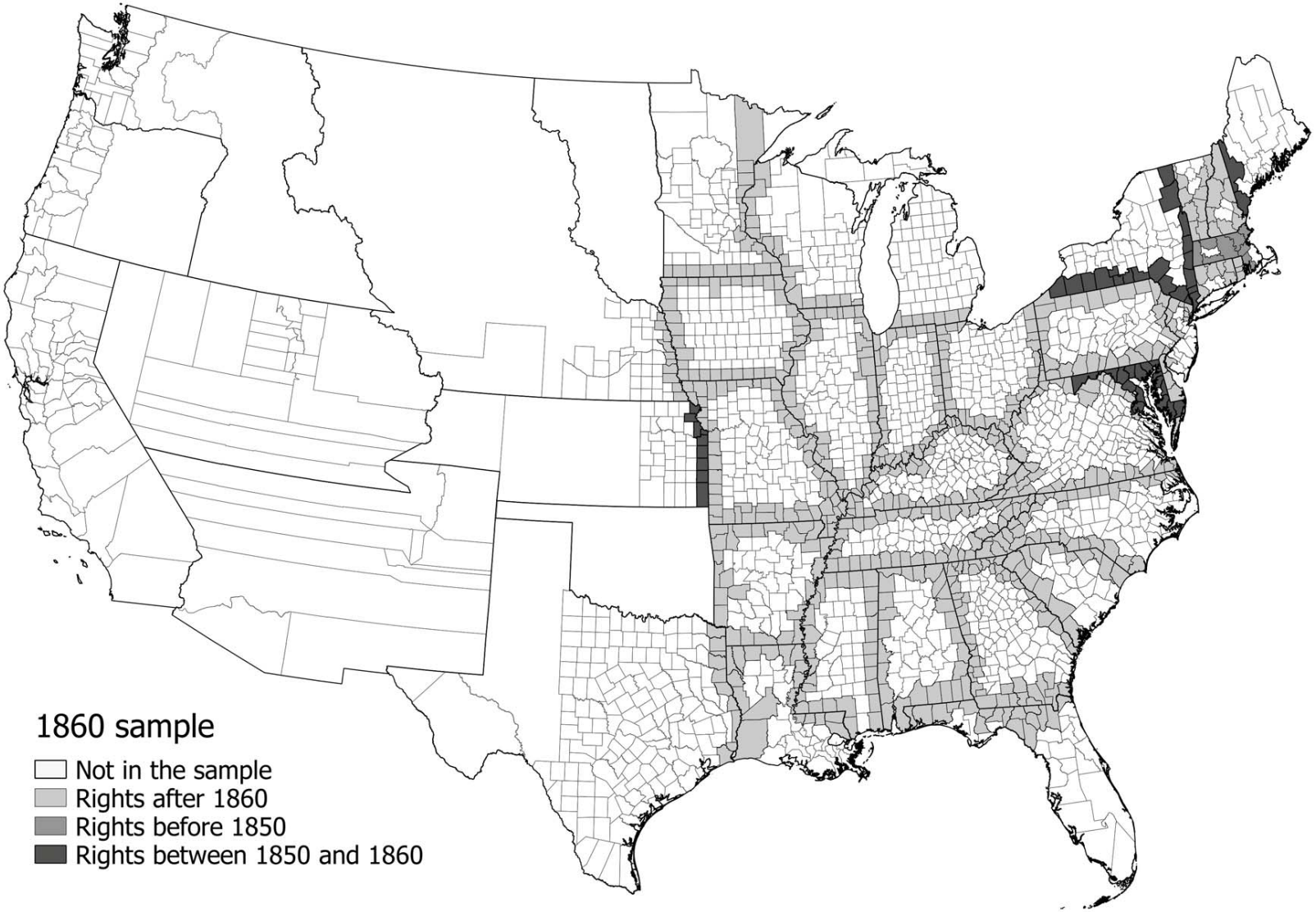
$$Y_{hsct} = \sum_k \alpha_k \cdot rights_{st}^k + \beta_{c,b(c)} + \gamma_{c,b(c)} + \lambda_s + X'_{hsct}\delta + \epsilon_{hsct}$$

- Y_{hsct}: dependent variable (married, fertility, LFP).
- $rights_{st}^k$: series of dummy variables set equal to one if a state had granted rights k years ago, where $k \in \{\leq -30, -20, -10, 0, 10, 20, \geq 30\}$.
- $\beta_{c,b(c)}$: fixed effects for each county c and its border pair $b(c)$.
- $\gamma_{c,b(c)}$ are linear time-trends for each county-border pair.
- λ_s are state fixed effects.
- X'_{hsct} controls: age, husband's age, occupation and industry FE, all interacted w. yr. FE

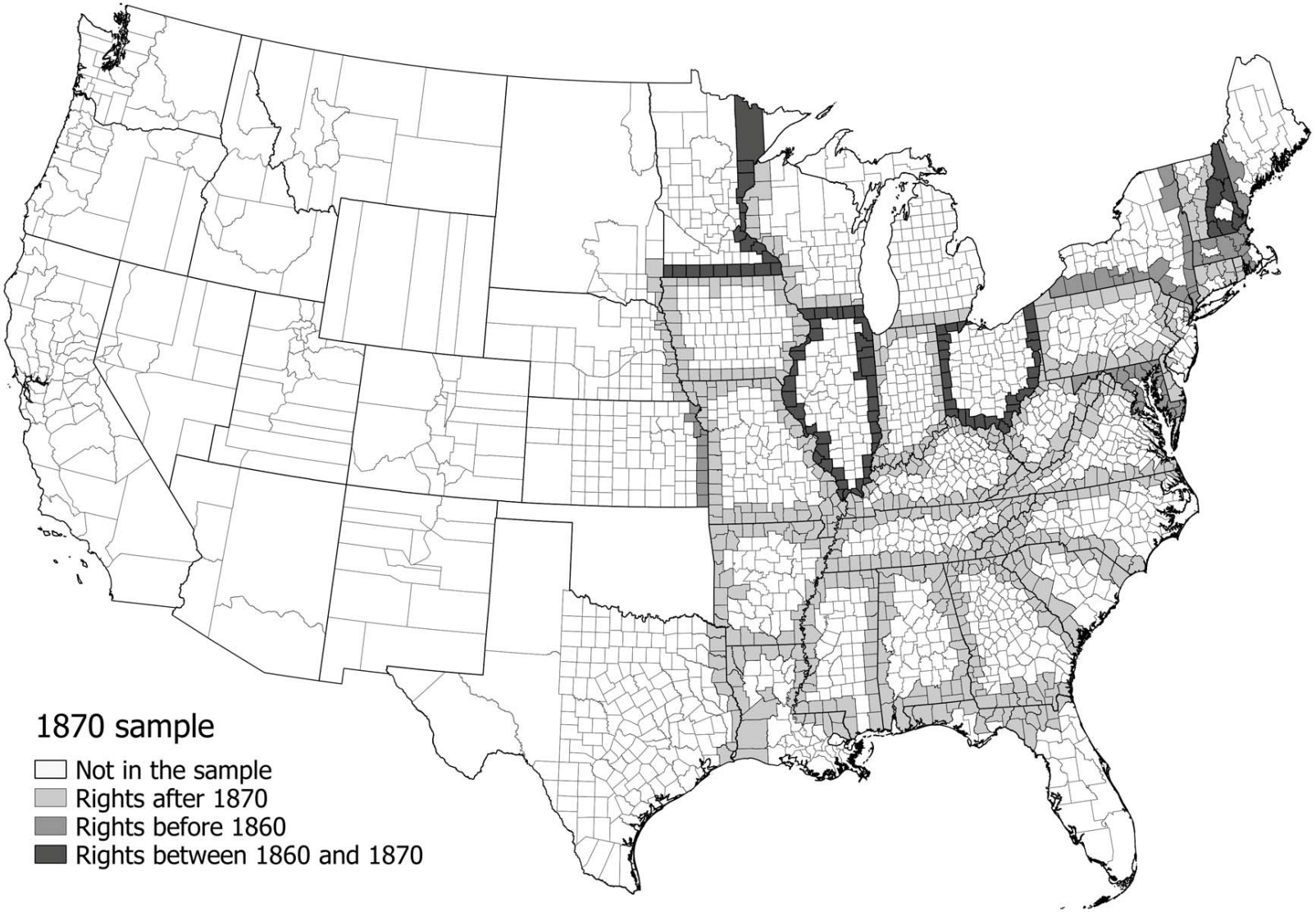
1850


1850 sample
Not in the sample
Rights after 1850
Rights before 1840
Rights between 1840 and 1850

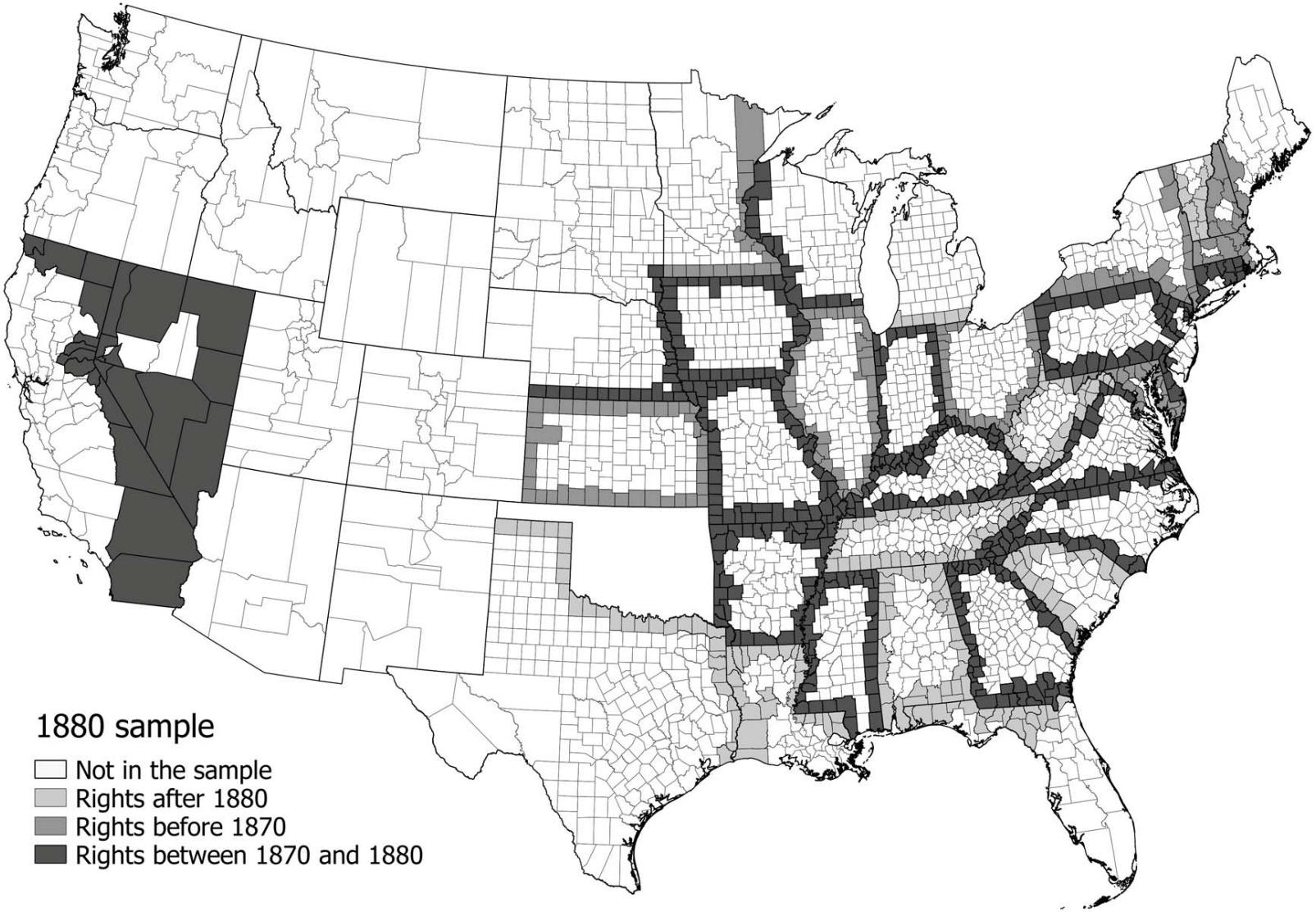
1860


1860 sample
Not in the sample
Rights after 1860
Rights before 1850
Rights between 1850 and 1860

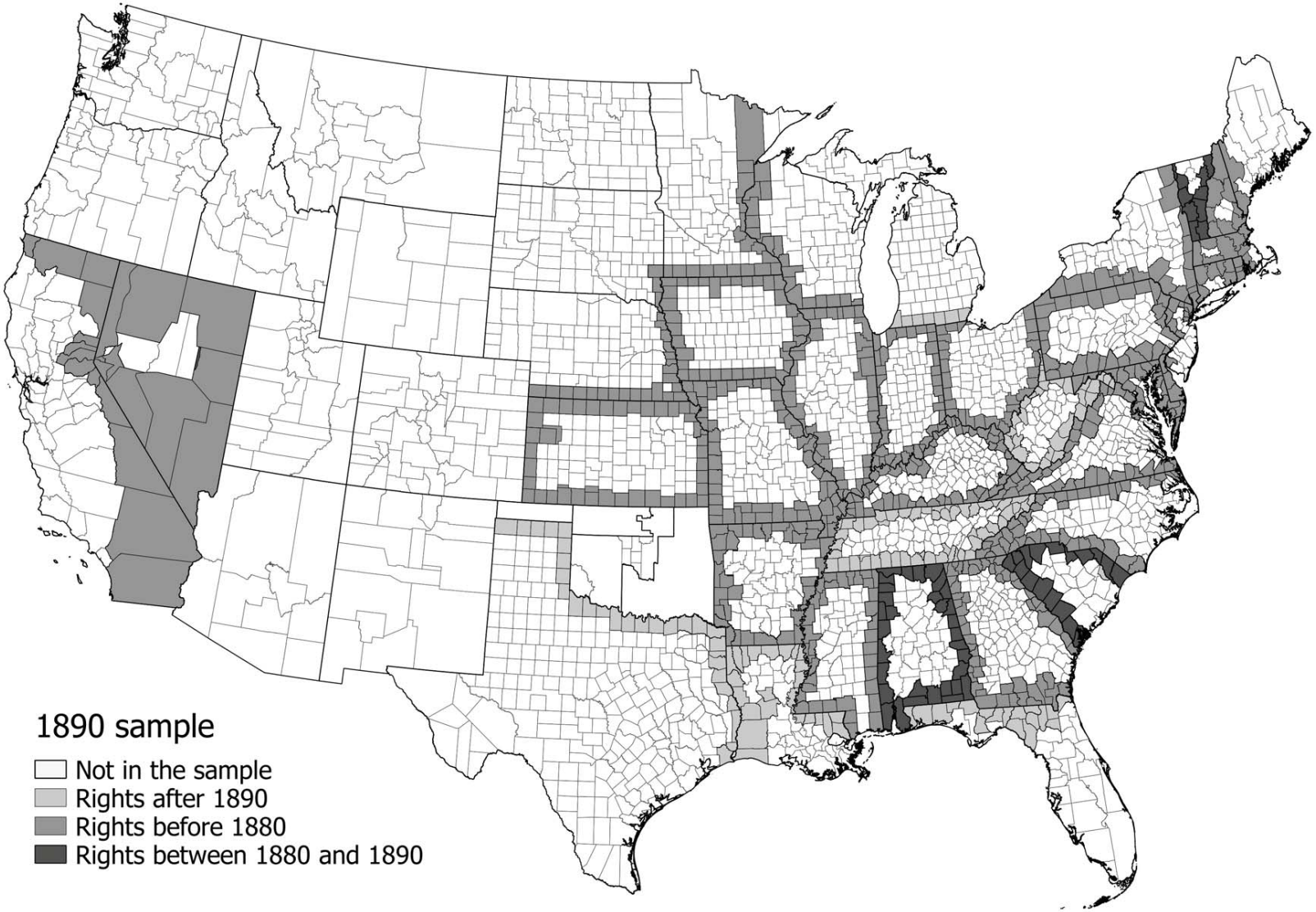
1870


1870 sample
Not in the sample
Rights after 1870
Rights before 1860
Rights between 1860 and 1870

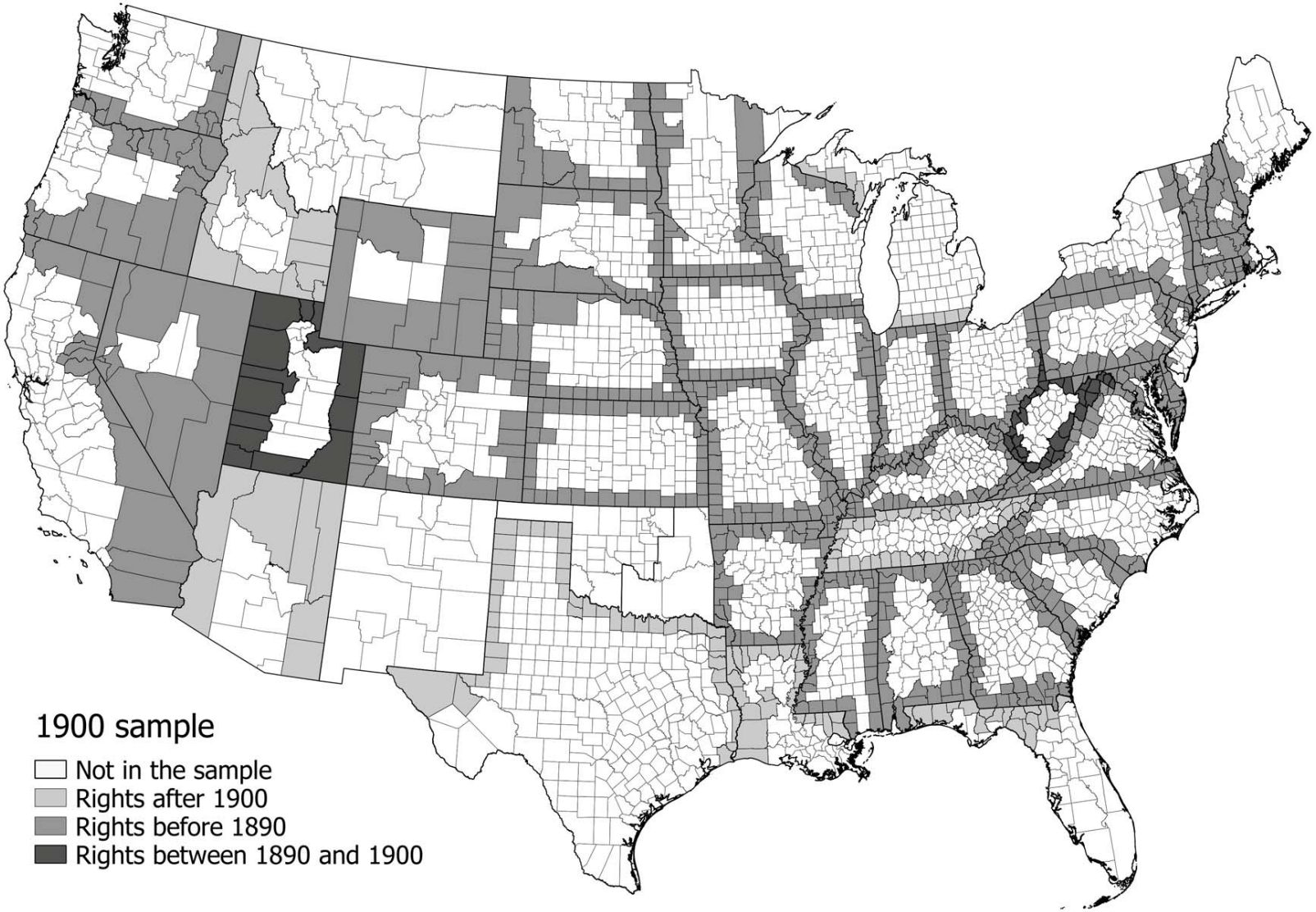
1880


1880 sample
Not in the sample
Rights after 1880
Rights before 1870
Rights between 1870 and 1880

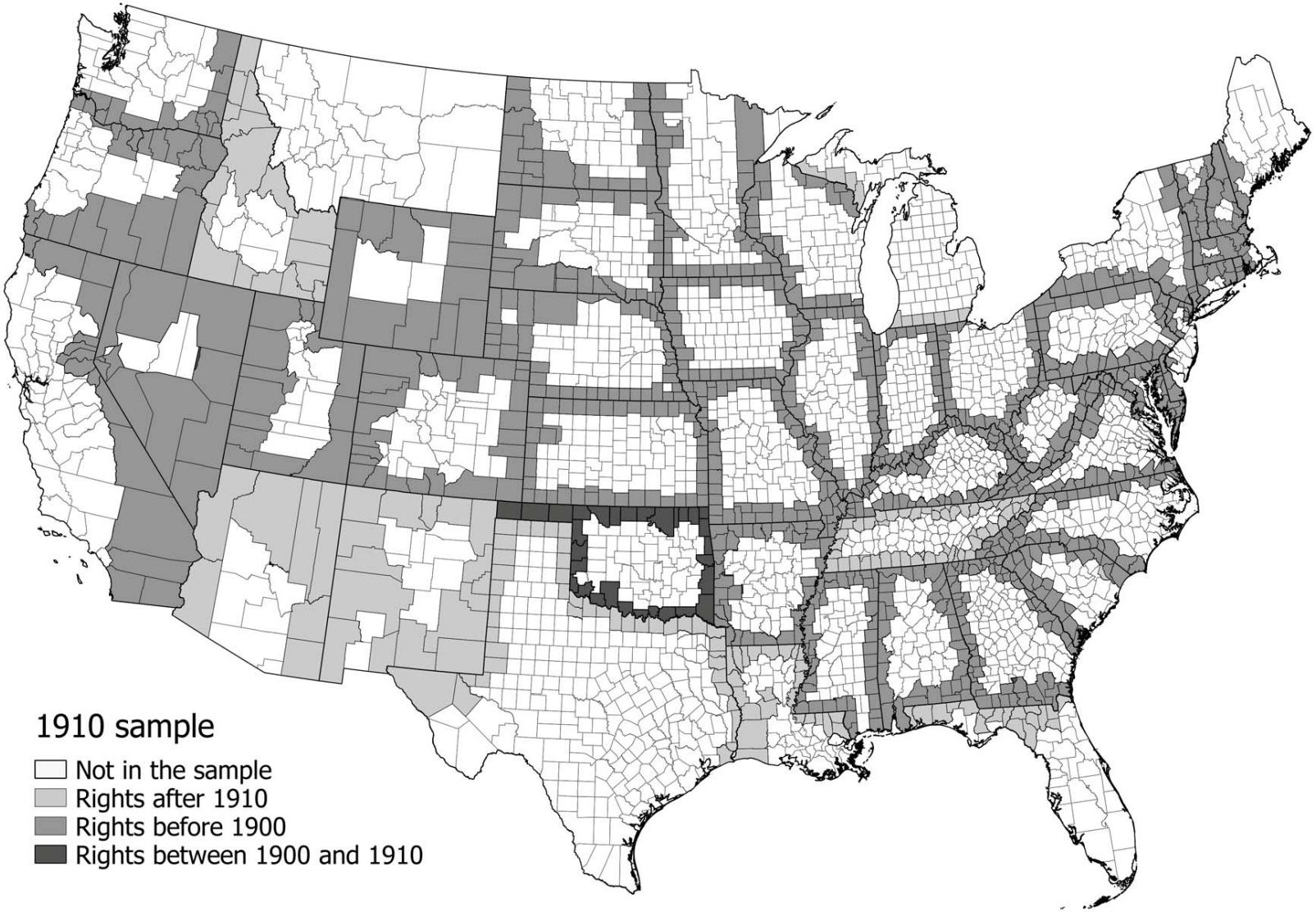
1890


1890 sample
Not in the sample
Rights after 1890
Rights before 1880
Rights between 1880 and 1890

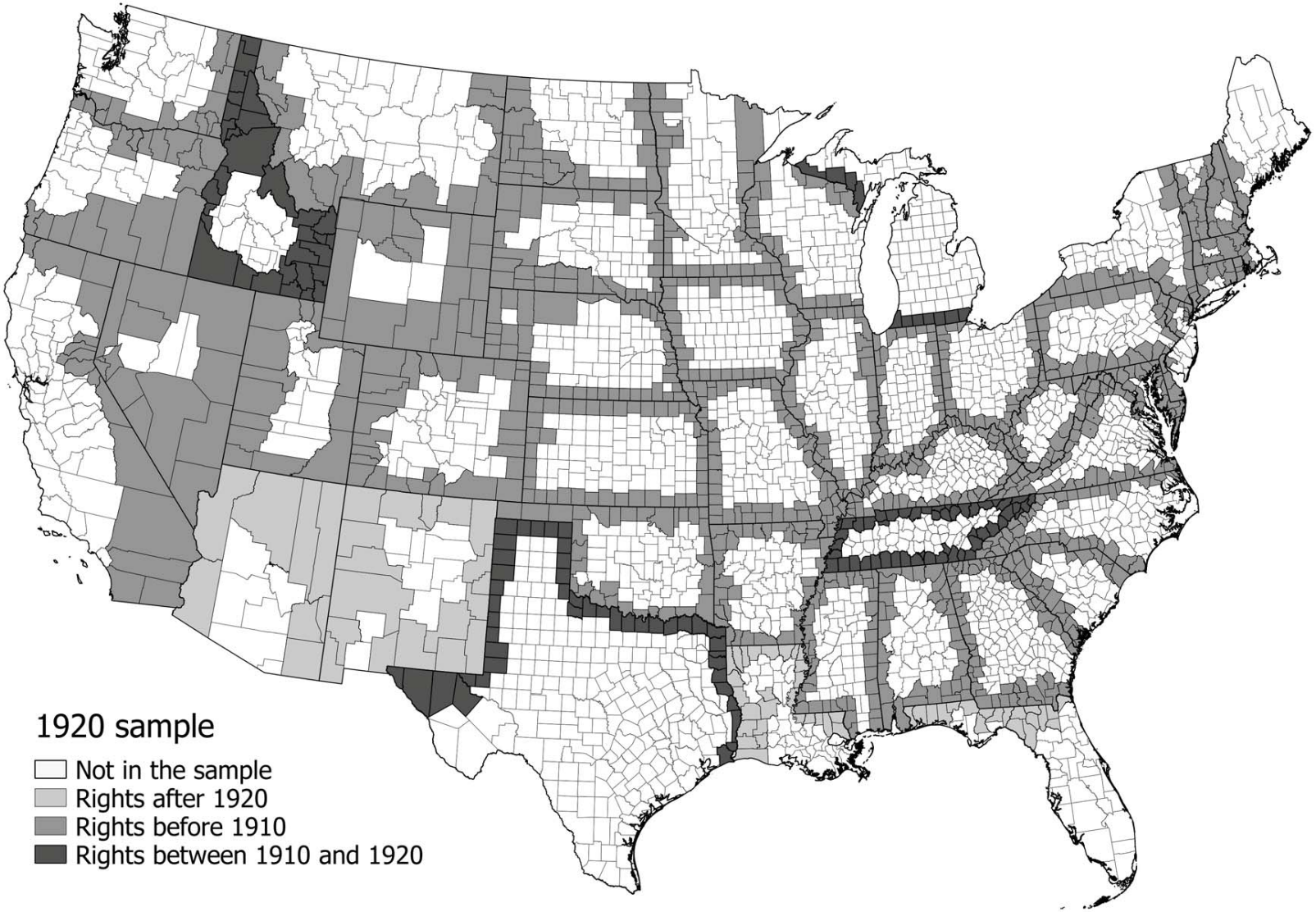
1900


1900 sample
Not in the sample
Rights after 1900
Rights before 1890
Rights between 1890 and 1900

1910


1910 sample
Not in the sample
Rights after 1910
Rights before 1900
Rights between 1900 and 1910

1920


1920 sample
Not in the sample
Rights after 1920
Rights before 1910
Rights between 1910 and 1920

Event Study: Discussion

- Issues building the maps
  - What is "a county"
  - A county may border more than one county.
    - Duplicate observations- reweight.
    - Double cluster: county-pair, state (Dube at al. 2010)
- Parallel trends
- Policy change exogenous to individual counties
  - Substantial heterogeneity between counties within states.
- No evidence of other contemporaneous policy changes.
- Robustness:
  - South: drop border South/non-South, Community Property.
  - Two Way FE issues in Diff in Diff: Use 2-step estimator as in Thakral (2021) and Gardner (2021)
    - Very similar. Use classic event study as benchmark.

Married Before/After Rights: 1900-1910

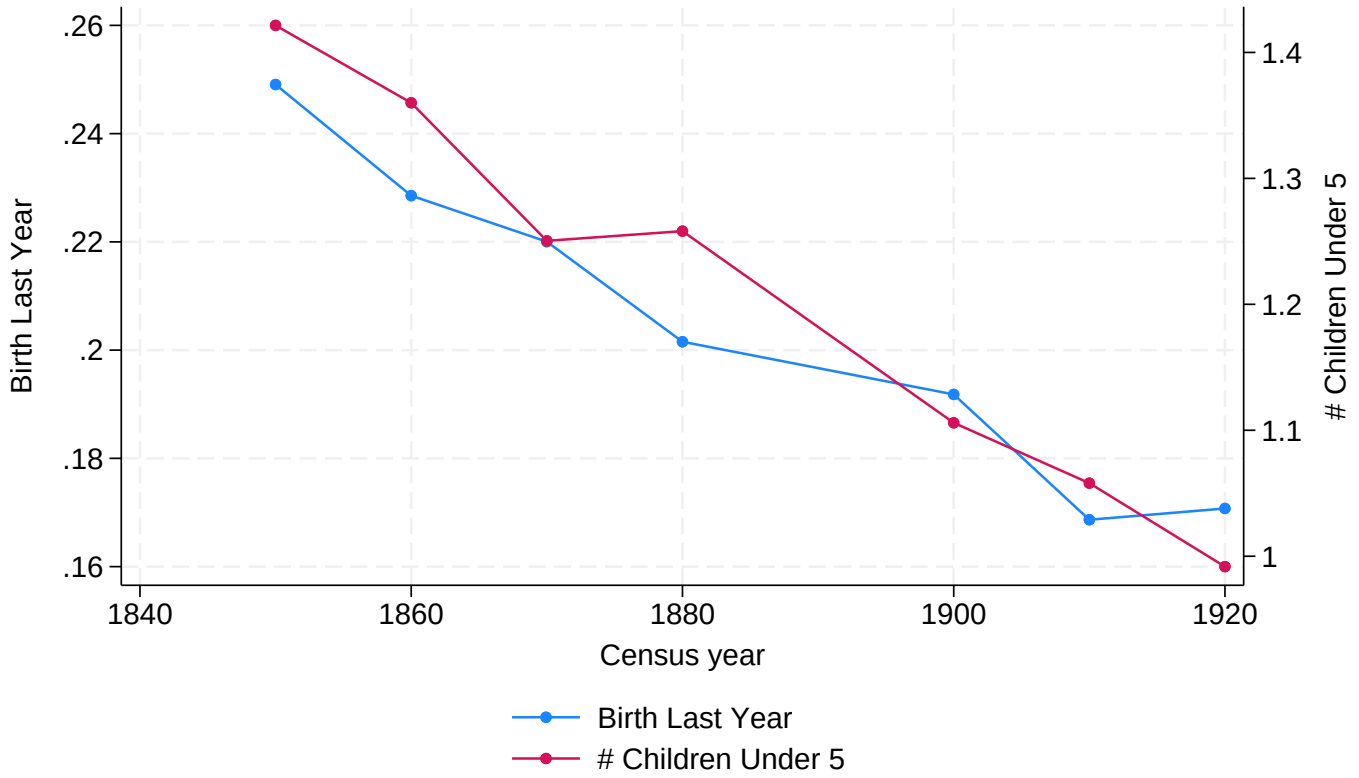
$$Y_{hsct} = \alpha \cdot Married\,Rights_{hsct} + \beta_{c,t} + X'_{hsct}\delta + \epsilon_{hsct}$$

- Y_{hsct}: dependent variable (fertility, LFP).
- $Married\ Rights_{hsct}$ takes the value of 1 if the couple was married with rights.
- $\beta_{c,t}$ are county-year fixed effects.
- X'_{hsct} controls variables.
- Standard errors clustered at the state level.

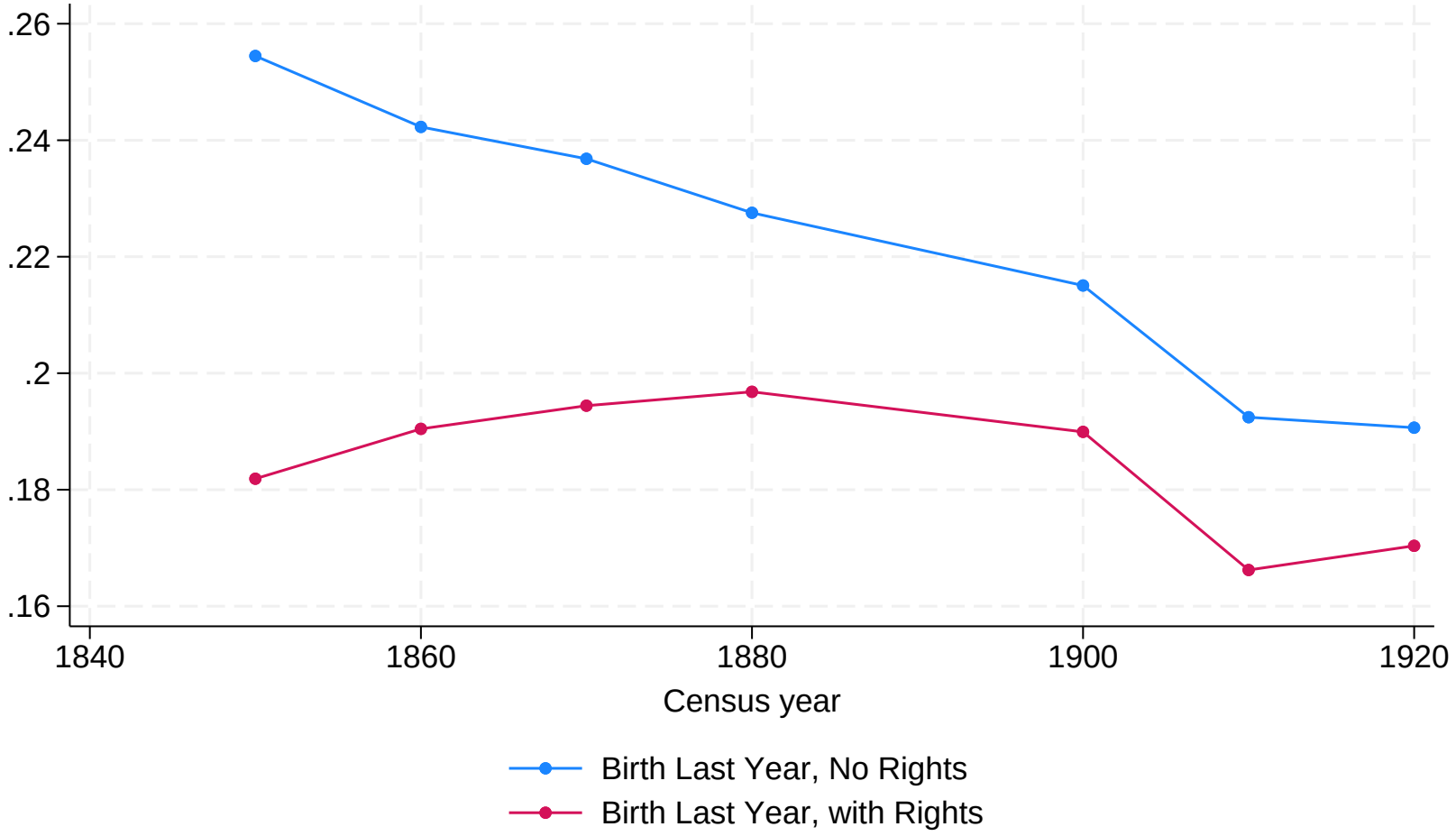
Marriage Market- Event Study

Fertility: 1850-1920

- Dependent variables:
  - Birth last year (Mean=0.2; 1850 mean=0.25; 1920 mean=0.17; $\Delta_{1850-1920} = 0.08$)
  - # Kids under 5 (Mean = 1.17; 1850 mean=1.42; 1920 mean=0.99; $\Delta_{1850-1920} = 0.43$)



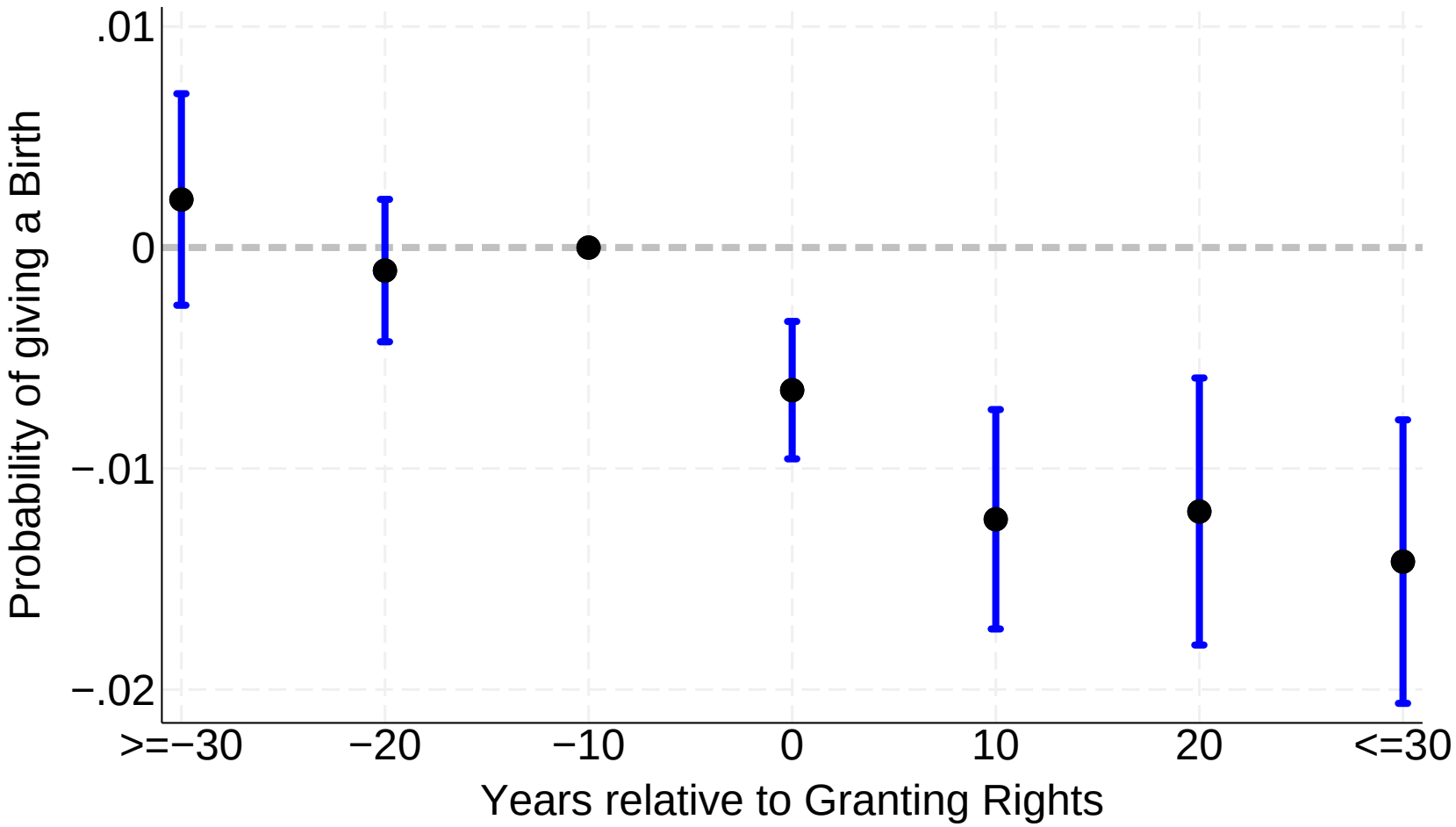
Birth Last Year by Rights: 1850-1920



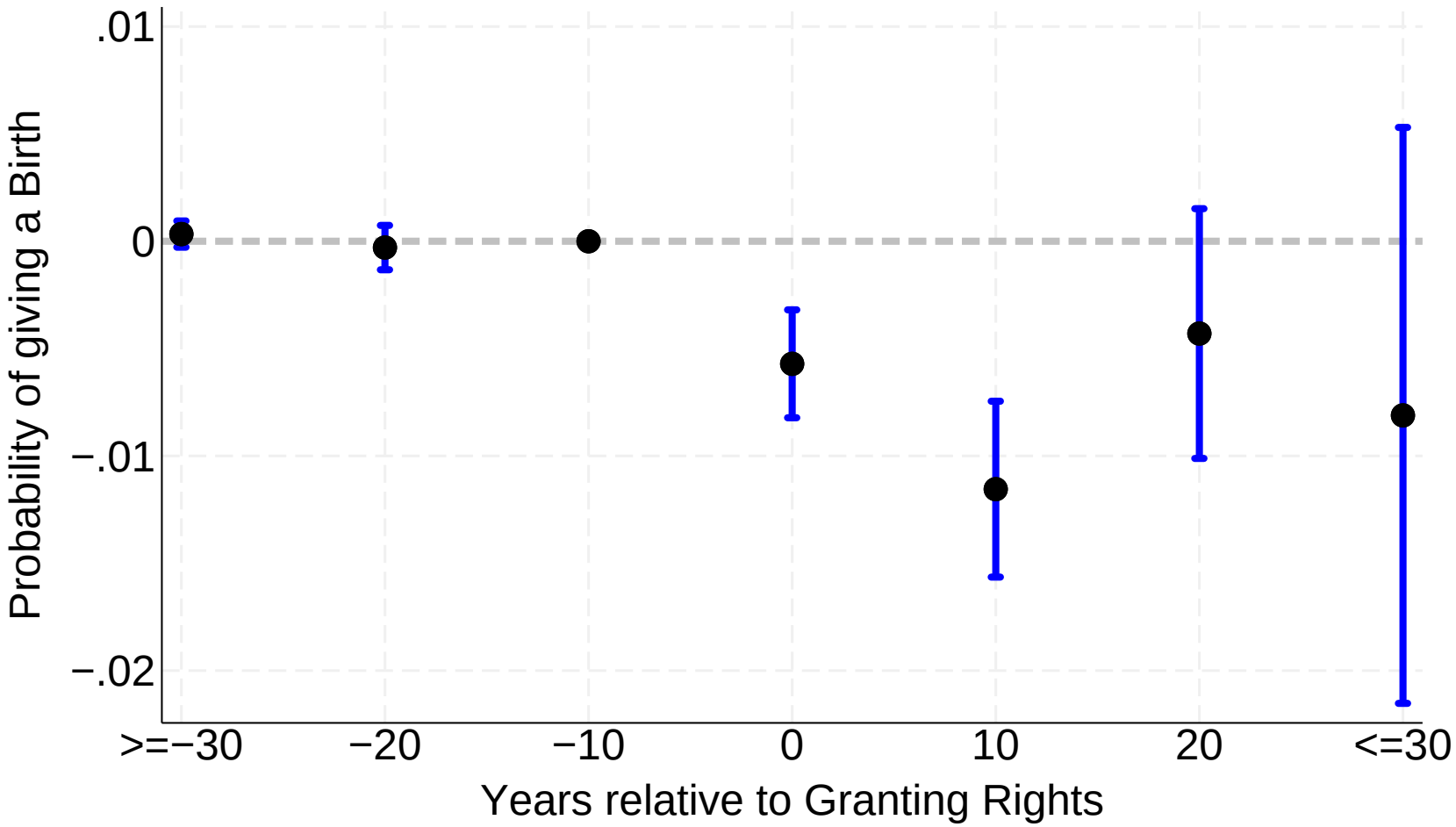
Probability of Giving Birth: 1850-1920

Dependent Variable:	Probability of Giving a Birth			
Sample	All US Born WNH, Women 20-39		Stayers, Women 20-39	
	(1)	(2)	(3)	(4)
$\geq$ 3 Decades Before	0.002	0.000	0.001	0.001***
	(0.002)	(0.000)	(0.005)	(0.000)
2 Decades Before	-0.001	-0.000	-0.001	0.000
	(0.002)	(0.001)	(0.003)	(0.001)
1 Decade Before	0	0	0	0
Rights Given	-0.006***	-0.006***	-0.007***	-0.008***
	(0.002)	(0.001)	(0.002)	(0.001)
[.35em] 1 Decade After	-0.012***	-0.012***	-0.011***	-0.012***
	(0.003)	(0.002)	(0.004)	(0.003)
2 Decades After	-0.012***	-0.004	-0.012***	-0.010***
	(0.003)	(0.003)	(0.004)	(0.004)
$\geq$ 3 Decades After	-0.014***	-0.008	-0.015***	-0.010**
	(0.003)	(0.007)	(0.004)	(0.005)
Controls	Yes	Yes	Yes	Yes
Extra Controls	Yes	Yes	Yes	Yes
		All-Two Step		All-Two Step
N	21,973,593	20,434,899	14,460,963	13,403,911
Adj. R^2	0.028	–	0.027	–
Mean Dep. Var.	0.198	0.194	0.20	0.20

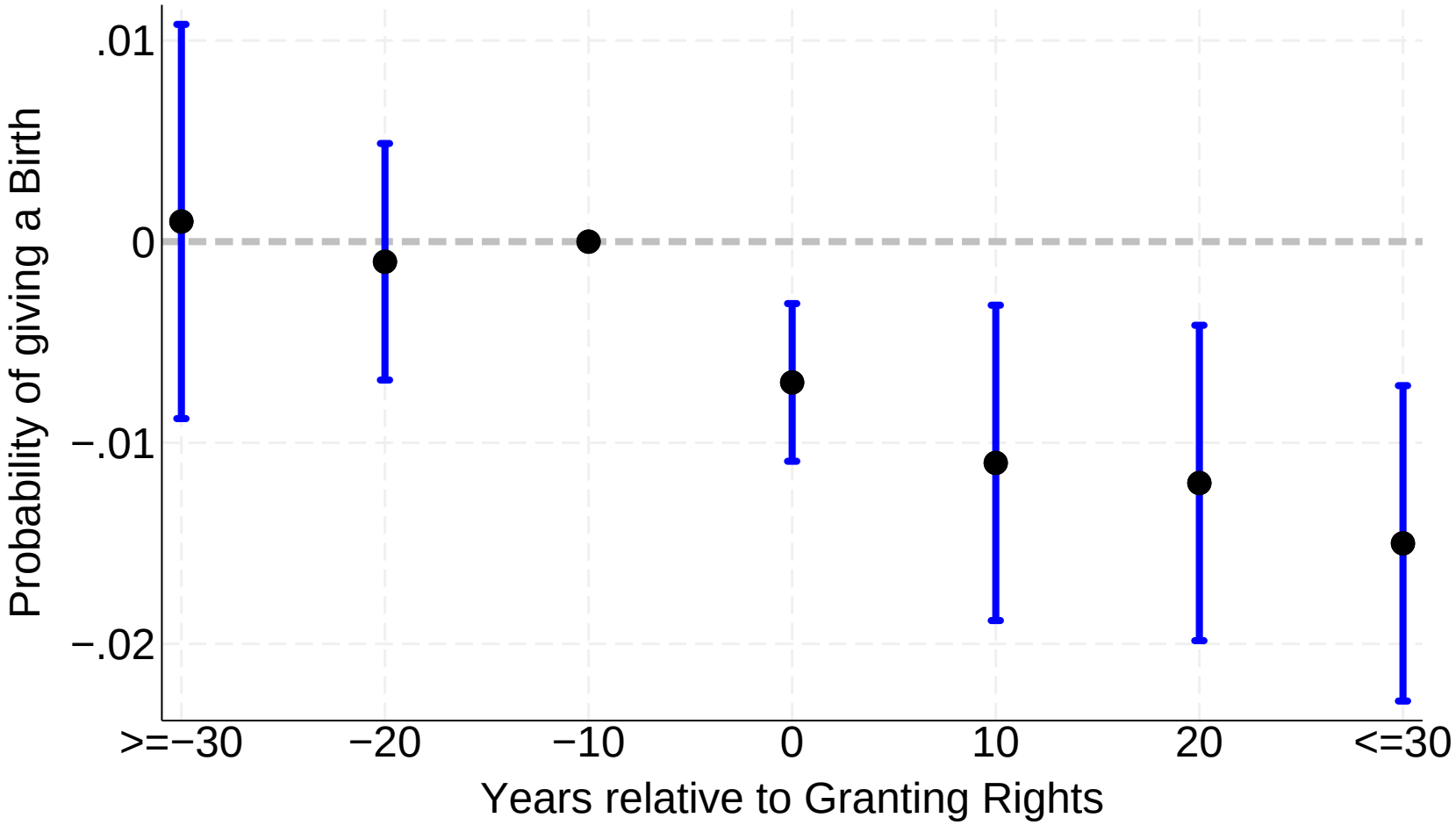
Probability of Giving Birth: 1850-1920



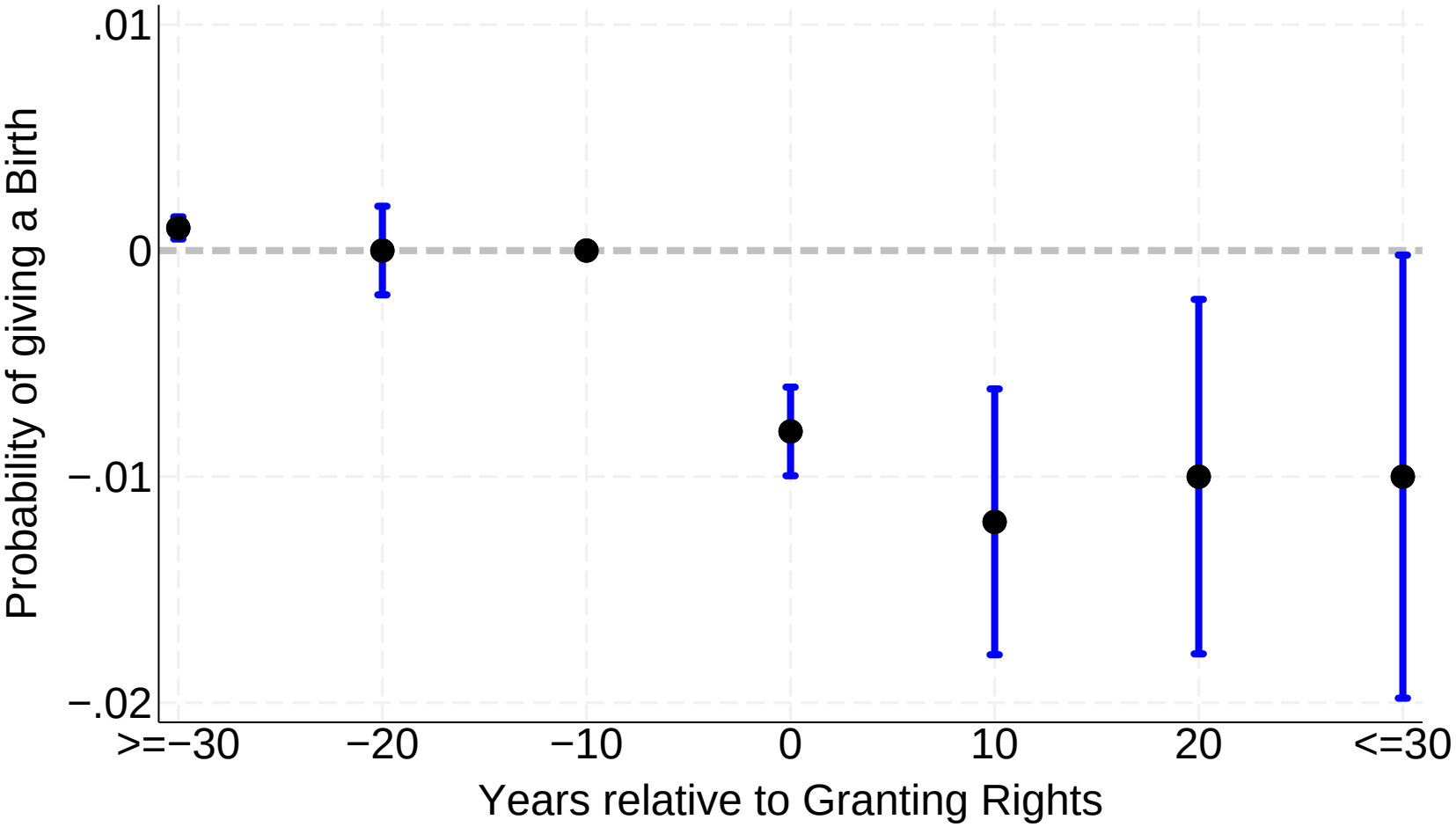
Prob. Giving Birth: 1850-1920 – Two Step Estimator



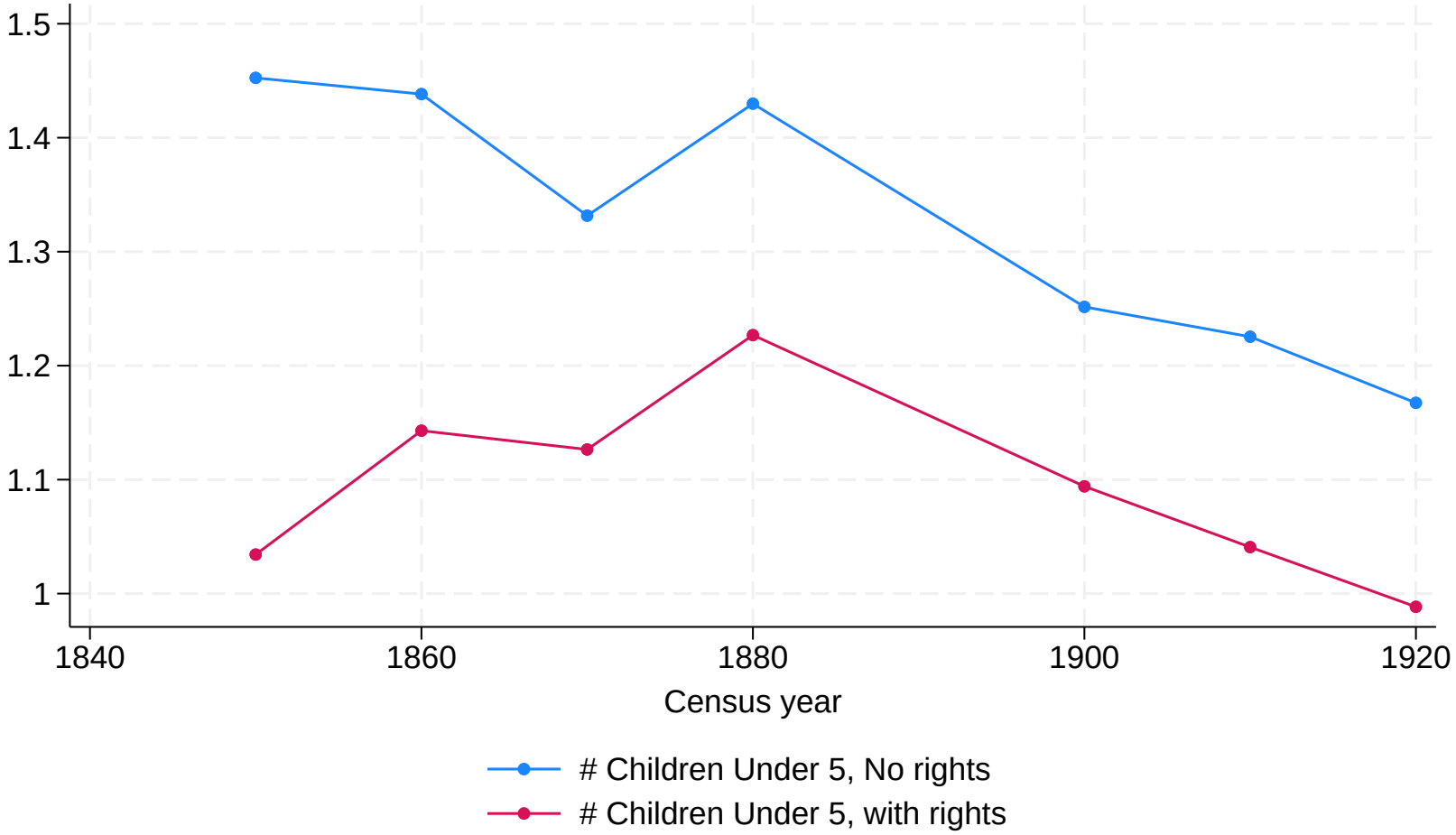
Robustness: Probability of Giving Birth: "Stayers", 1850-1920



Robustness: Prob. Giving Birth: "Stayers", 1850-1920 – Two Step Estimator



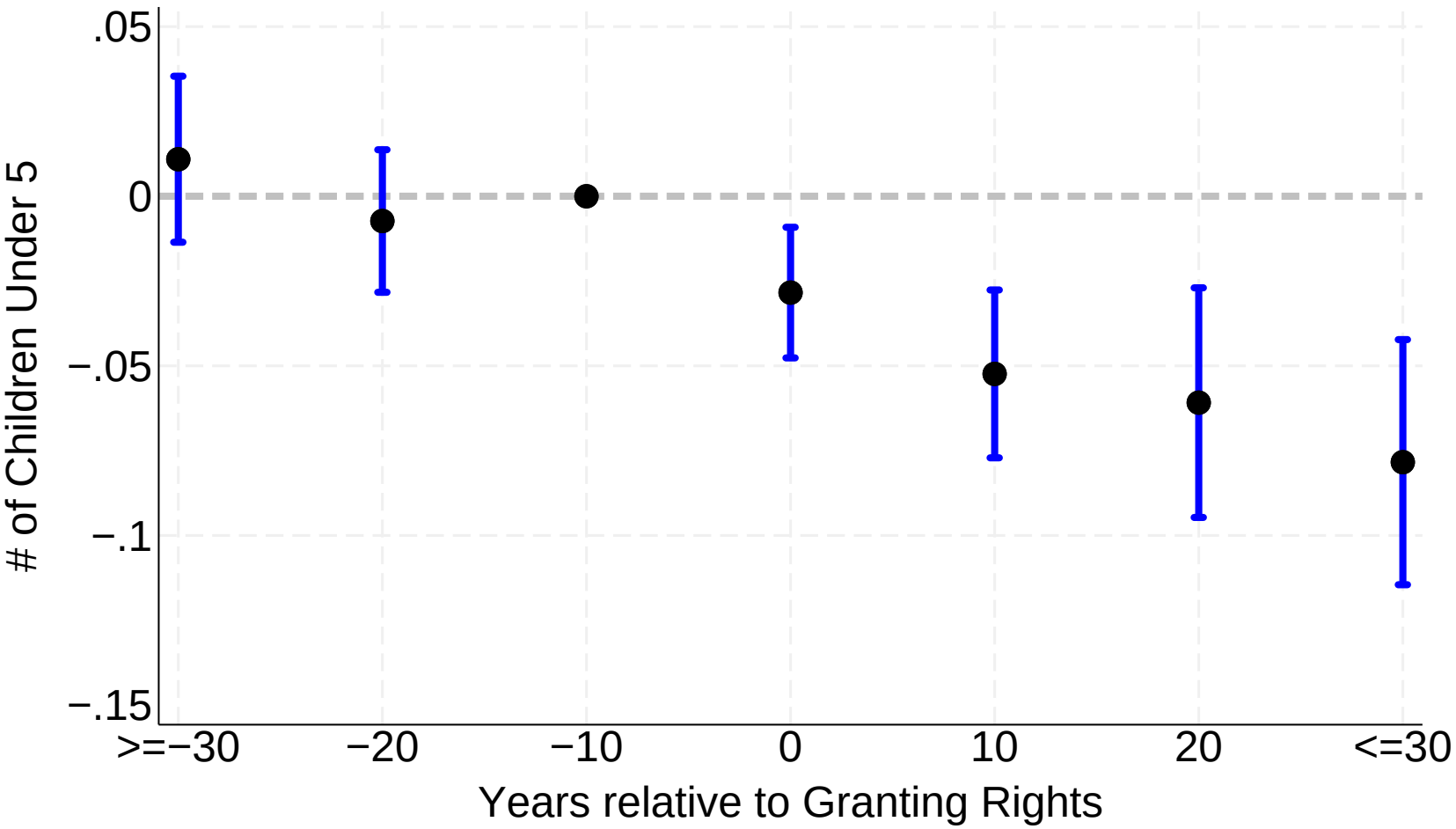
Number of Children Under Age 5: 1850-1920



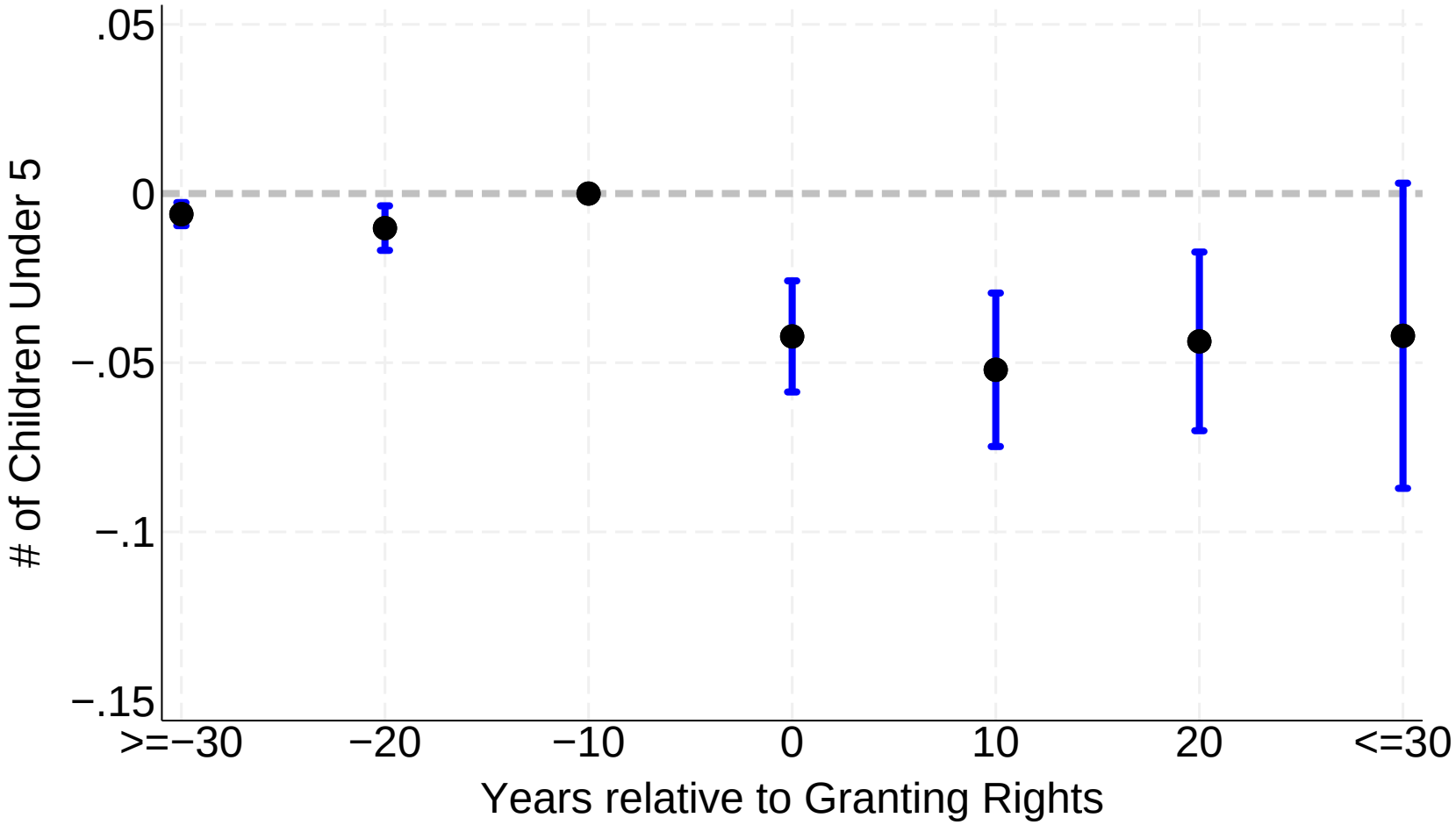
Number of Children under Age 5: 1850-1920

Dependent Variable:	Number of Children Under Age 5			
Sample	All US Born WNH, Women 20-39		Stayers, Women 20-39	
	(1)	(2)	(3)	(4)
$\geq$ 3 Decades Before	0.011	-0.006***	0.008	-0.005***
	(0.012)	(0.000)	(0.020)	(0.002)
2 Decades Before	-0.007	-0.010***	-0.014	-0.013***
	(0.011)	(0.003)	(0.016)	(0.004)
1 Decade Before	0	0	0	0
Rights Given	-0.028***	-0.042***	-0.029*	-0.053***
	(0.010)	(0.008)	(0.016)	(0.012)
1 Decade After	-0.052***	-0.052***	-0.040**	-0.047***
	(0.013)	(0.012)	(0.017)	(0.015)
2 Decades After	-0.061***	-0.044***	-0.059***	-0.068***
	(0.017)	(0.013)	(0.021)	(0.017)
$\geq$ 3 Decades After	-0.078***	-0.042*	-0.081***	-0.076***
	(0.018)	(0.023)	(0.023)	(0.024)
Controls	Yes	Yes	Yes	Yes
Extra Controls	Yes	Yes	Yes	Yes
	All	All-Two Step	All	All-Two Step
N	21,973,593	20,434,899	14,460,963	13,403,911
Adj. R^2	0.124	–	0.120	–
Mean Dep. Var.	1.173	1.157	1.19	1.19

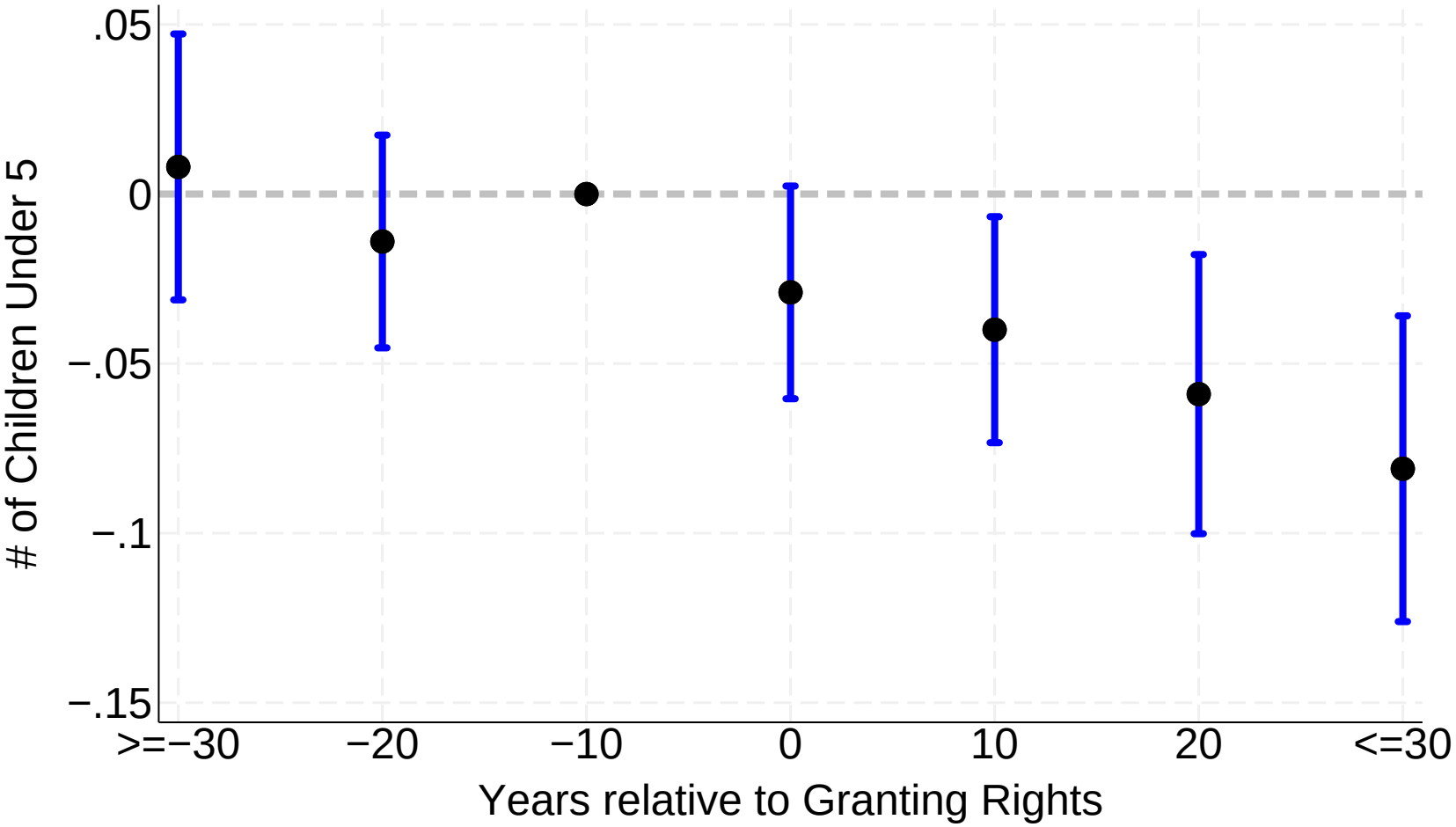
Number of Children under Age 5: 1850-1920



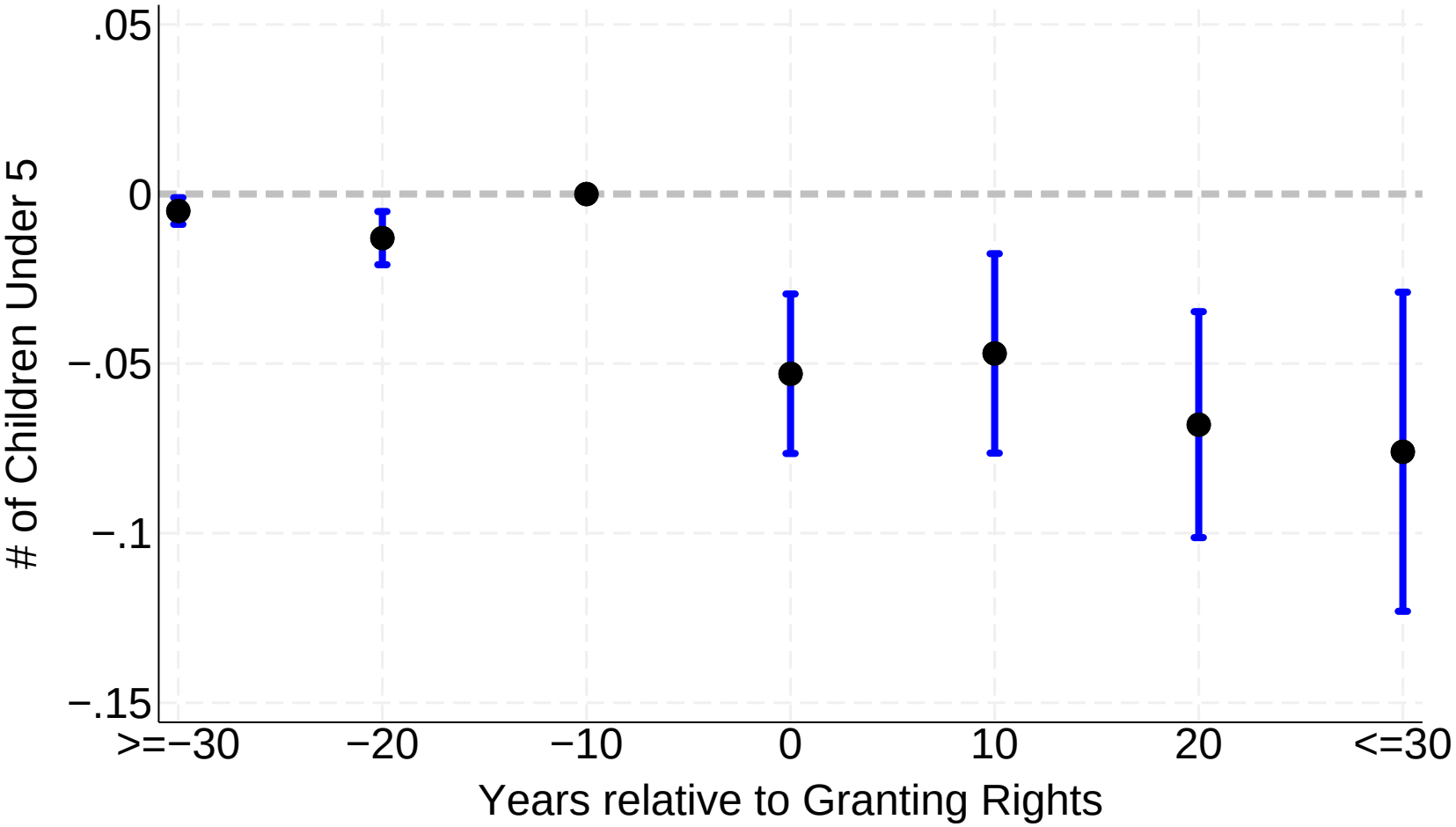
Number of Children under Age 5: 1850-1920 – Two Step Estimator



Robustness: Number of Children under Age 5: "Stayers", 1850-1920



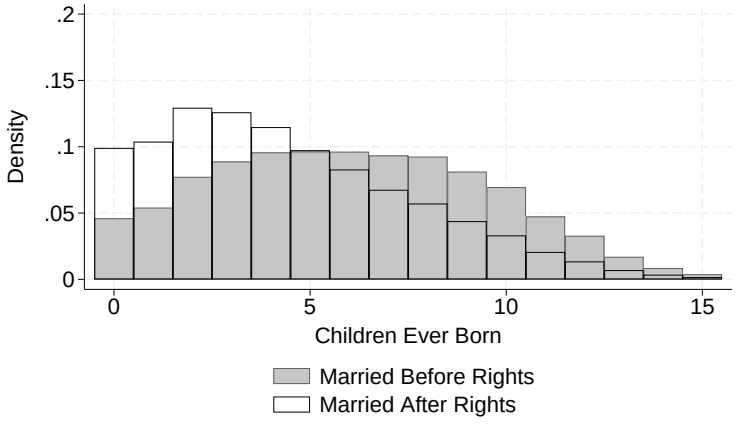
Robustness: Number of Children under Age 5: "Stayers", 1850-1920 – Two Step Estimator



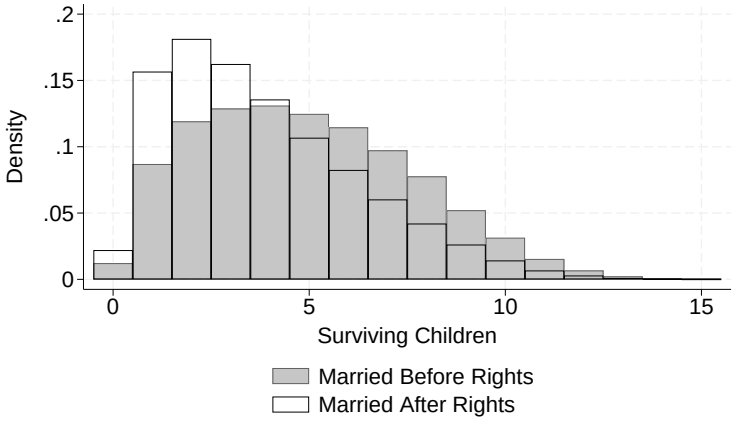
Robustness: Flow Measures of Fertility, 1850-1920

Dependent Variable	Birth Last Year			# Children Under 5		
	(1)	(2)	(3)	(4)	(5)	(6)
	No Extra	No South	No CP	No Extra	No South	No CP
$\geq$ 3 Decades Before	0.002	0.002	0.002	0.010	0.017	0.011
	(0.003)	(0.003)	(0.003)	(0.015)	(0.013)	(0.013)
2 Decades Before	-0.001	-0.001	-0.001	-0.007	-0.006	-0.007
	(0.002)	(0.002)	(0.002)	(0.012)	(0.012)	(0.011)
1 Decade Before	0	0	0	0	0	0
Rights Given	-0.006***	-0.006***	-0.006***	-0.028***	-0.026***	-0.027***
	(0.002)	(0.002)	(0.002)	(0.010)	(0.010)	(0.010)
1 Decade After	-0.013***	-0.013***	-0.012***	-0.053***	-0.059***	-0.051***
	(0.003)	(0.003)	(0.003)	(0.014)	(0.012)	(0.013)
2 Decades After	-0.012***	-0.013***	-0.011***	-0.062***	-0.065***	-0.058***
	(0.003)	(0.003)	(0.003)	(0.017)	(0.016)	(0.018)
$\geq$ 3 Decades After	-0.015***	-0.016***	-0.013***	-0.084***	-0.084***	-0.076***
	(0.003)	(0.003)	(0.003)	(0.019)	(0.018)	(0.020)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Extra Controls	No	Yes	Yes	No	Yes	Yes
N	21,973,593	17,926,262	20,624,476	21,973,593	17,926,262	20,624,476
Adj. R^2	0.026	0.029	0.028	0.110	0.125	0.123
Mean Dep. Var.	0.198	0.194	0.198	1.173	1.152	1.174

Married-After: Completed Fertility & Surviving Children: 45-59, 1900-1910



Children Ever Born



Surviving Children

Married-After Fertility: 1900-1910. 20-39 and 45-59 yr old women

Dep Var:	(1) Birth Last Year	(2) #Children Under 5	(3) CEB	(4) CEB>0	(5) Surviving Children	(6) Surviving Ratio
	Women Age 20-39		Women Age 45-59			
Married After Rights	-0.009***	-0.116***	-0.189***	-0.183***	-0.143***	-0.001
	(0.002)	(0.025)	(0.059)	(0.054)	(0.045)	(0.001)
N	10,040,159	10,040,159	3,485,362	3,191,435	3,191,435	3,191,435
Adj. R^2	0.052	0.201	0.261	0.234	0.213	0.027
Mean Dep. Var.	0.180	1.072	4.844	5.290	4.146	0.807

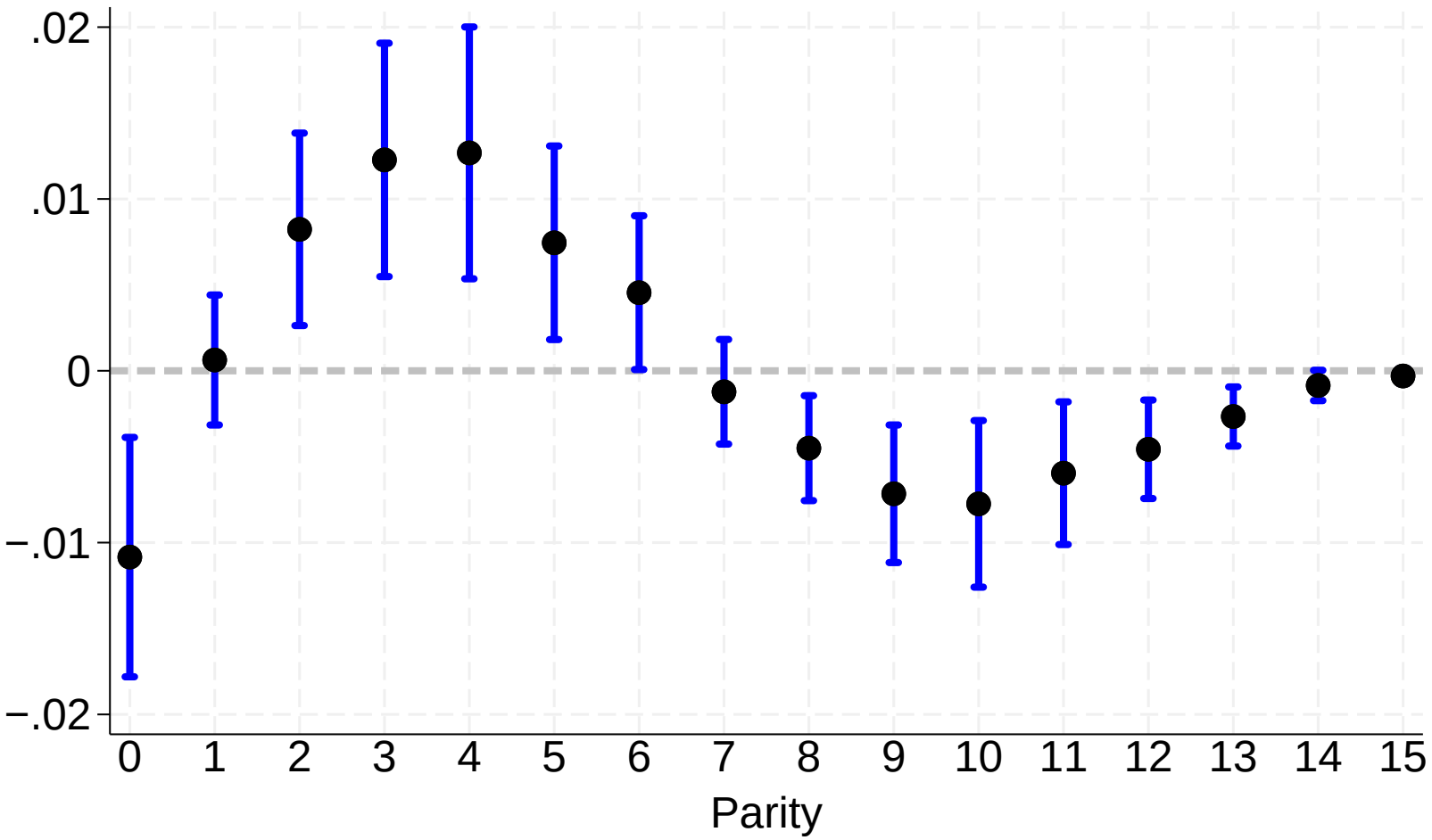
Robustness: Married-After Fertility: 1900-1910. 20-39 yr old women

	(1)	(2)	(3)	(4)
Sample	No Extra	No South	No CP	Donut
	Panel A: Dep Var: Birth Last year, age 20-39			
Married After	-0.010***	-0.010**	-0.009***	-0.010***
Rights	(0.002)	(0.004)	(0.002)	(0.002)
N	10,040,168	7,171,256	9,060,253	9,969,410
Adj. R^2	0.050	0.051	0.052	0.052
Mean Dep. Var.	0.180	0.164	0.179	0.180
	Panel B: Dep Var: # of Kids Under Age 5, age 20-39			
Married After	-0.121***	-0.114**	-0.121***	-0.132***
Rights	(0.024)	(0.051)	(0.027)	(0.026)
N	10,040,168	7,171,256	9,060,253	9,969,410
Adj. R^2	0.187	0.178	0.197	0.200
Mean Dep. Var.	1.072	0.975	1.065	1.071

Robustness: Married-After Fertility: 1900-1910. 45-59 yr old women

Sample	(1) All	(2) No South	(3) No CP	(4) Donut
	Panel A: Dep Var: Children Ever Born, age 45-59			
Married After Rights	-0.186***	-0.155***	-0.202***	-0.238***
	(0.058)	(0.055)	(0.062)	(0.071)
N	3,485,383	2,537,399	3,212,896	3,260,133
Adj. R^2	0.240	0.218	0.254	0.261
Mean Dep. Var.	4.844	4.391	4.789	4.777
	Panel B: Dep Var: Surviving Children, age 45-59			
Married After Rights	-0.146***	-0.104**	-0.156***	-0.184***
	(0.049)	(0.045)	(0.052)	(0.059)
N	3,485,383	2,537,399	3,212,896	3,260,133
Adj. R^2	0.222	0.201	0.236	0.243
Mean Dep. Var.	3.796	3.447	3.756	3.745

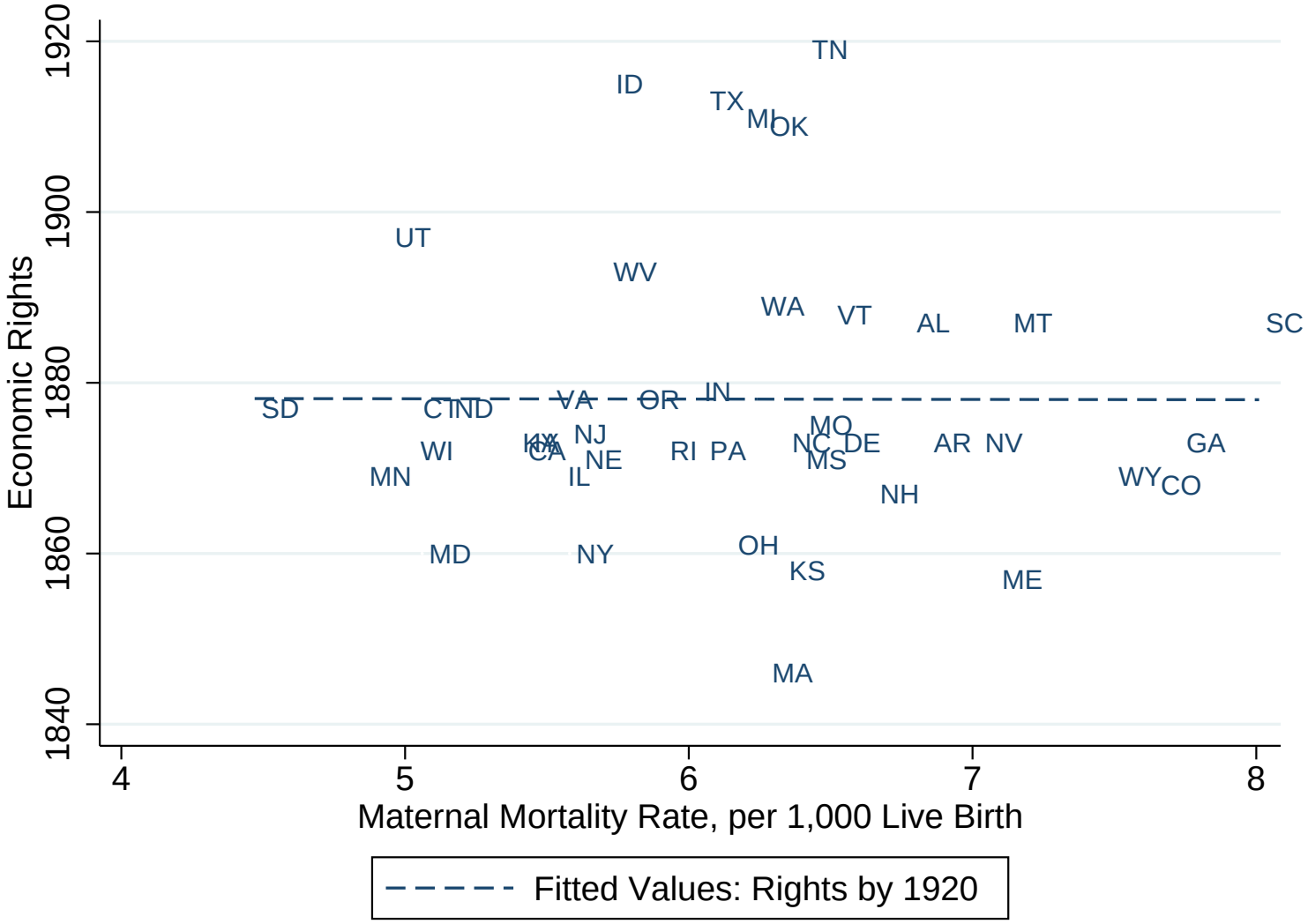
Married-After: CEB by Parity. 45-59, 1900-1910



Mechanisms

- Shift in bargaining power within the household is consistent with:
  1. Quantitatively, before/after estimates similar to event study estimates
  2. Suggestive evidence: Maternal Mortality Risk
  3. Suggest evidence: Wealth
  4. LFP did not change
- Notice: bargaining need not depend on threat of divorce, but on a noncooperative equilibrium. (Lundberg and Pollak, 1993)

Maternal Mortality Rate and Rights



Birth Last Year & #of Kids Under Age 5, Married After Rights 1900-1910 by **Maternal Mortality Rate States**

Dependent Variable:	Panel A: Women Age 20-39					
	Birth Last Year			# Children Under 5		
	Baseline	(2)	(3)	Baseline	(5)	(6)
Married After Rights	-0.009***	-0.006***	-0.008***	-0.116***	-0.098***	-0.106***
	(0.002)	(0.002)	(0.002)	(0.025)	(0.021)	(0.022)
Married After Rights		-0.009***	-0.005*		-0.067***	-0.047***
×High MMR		(0.003)	(0.004)		(0.018)	(0.016)
Sample	All	All	Rights$\leq$ 1920	All	All	Rights$\leq$ 1920
N	10,040,159	10,040,159	9,856,715	10,040,159	10,040,159	9,856,715
Adj. R^2	0.052	0.052	0.050	0.192	0.201	0.200
Mean Dep. Var.	0.180	0.180	0.178	1.070	1.072	1.067

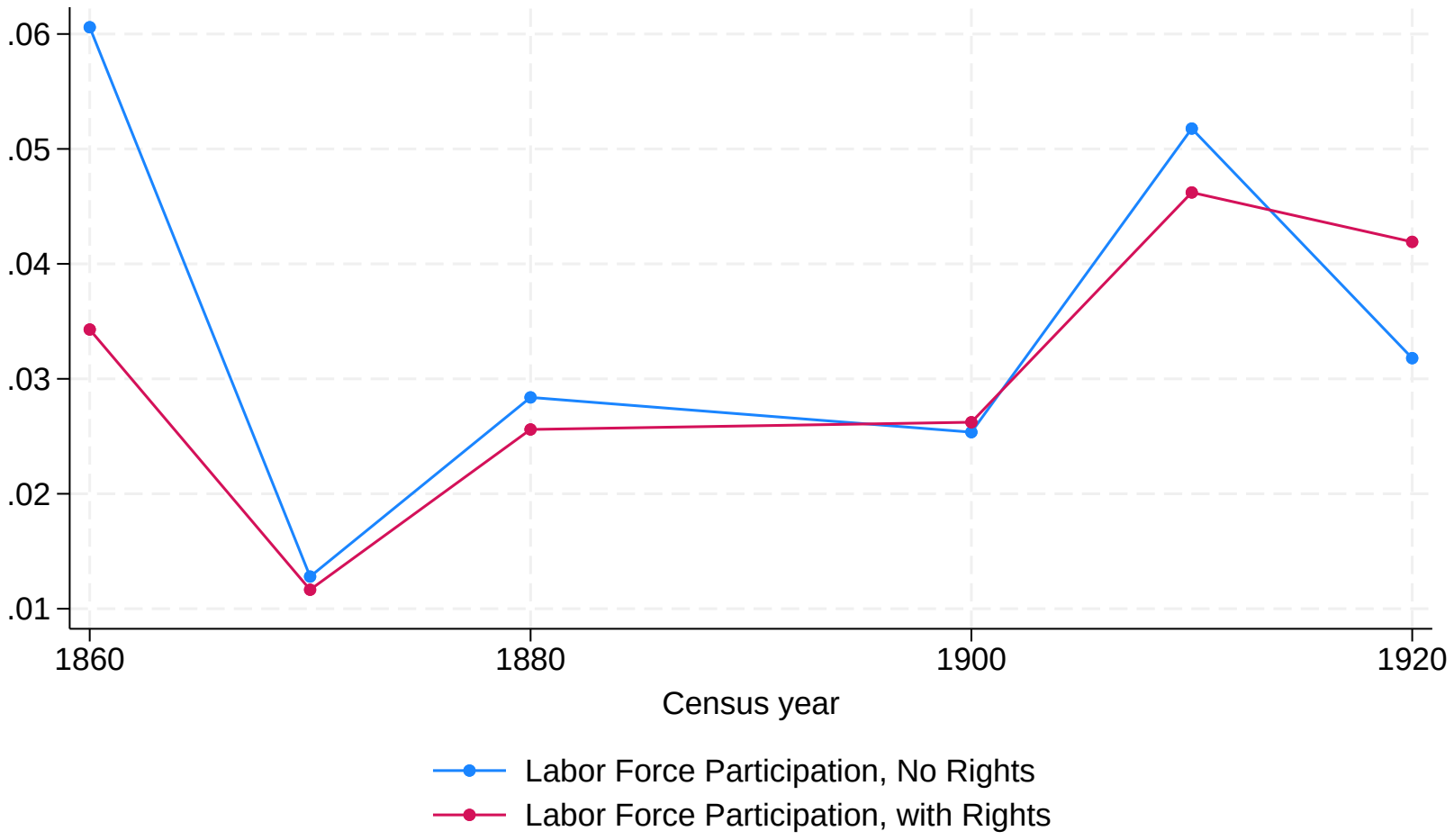
Children Ever Born & Surviving Children, Married After Rights 1900-1910 by **Maternal Mortality Rate States**

Dependent Variable:	Panel B: Women Age 45-59					
	Children Ever Born			Surviving Children		
	Baseline	(2)	(3)	Baseline	(5)	(6)
Married After Rights	-0.189***	-0.144**	-0.154***	-0.147***	-0.110**	-0.116**
	(0.059)	(0.055)	(0.055)	(0.049)	(0.045)	(0.045)
Married After Rights		-0.361***	-0.339**		-0.305***	-0.284**
×High MMR		(0.117)	(0.121)		(0.110)	(0.114)
Sample	All	All	Rights≤ 1920	All	All	Rights≤ 1920
N	3,485,362	3,485,362	3,438,518	3,485,362	3,485,362	3,438,518
Adj. R^2	0.261	0.261	0.259	0.242	0.242	0.241
Mean Dep. Var.	4.844	4.844	4.821	3.796	3.796	3.782

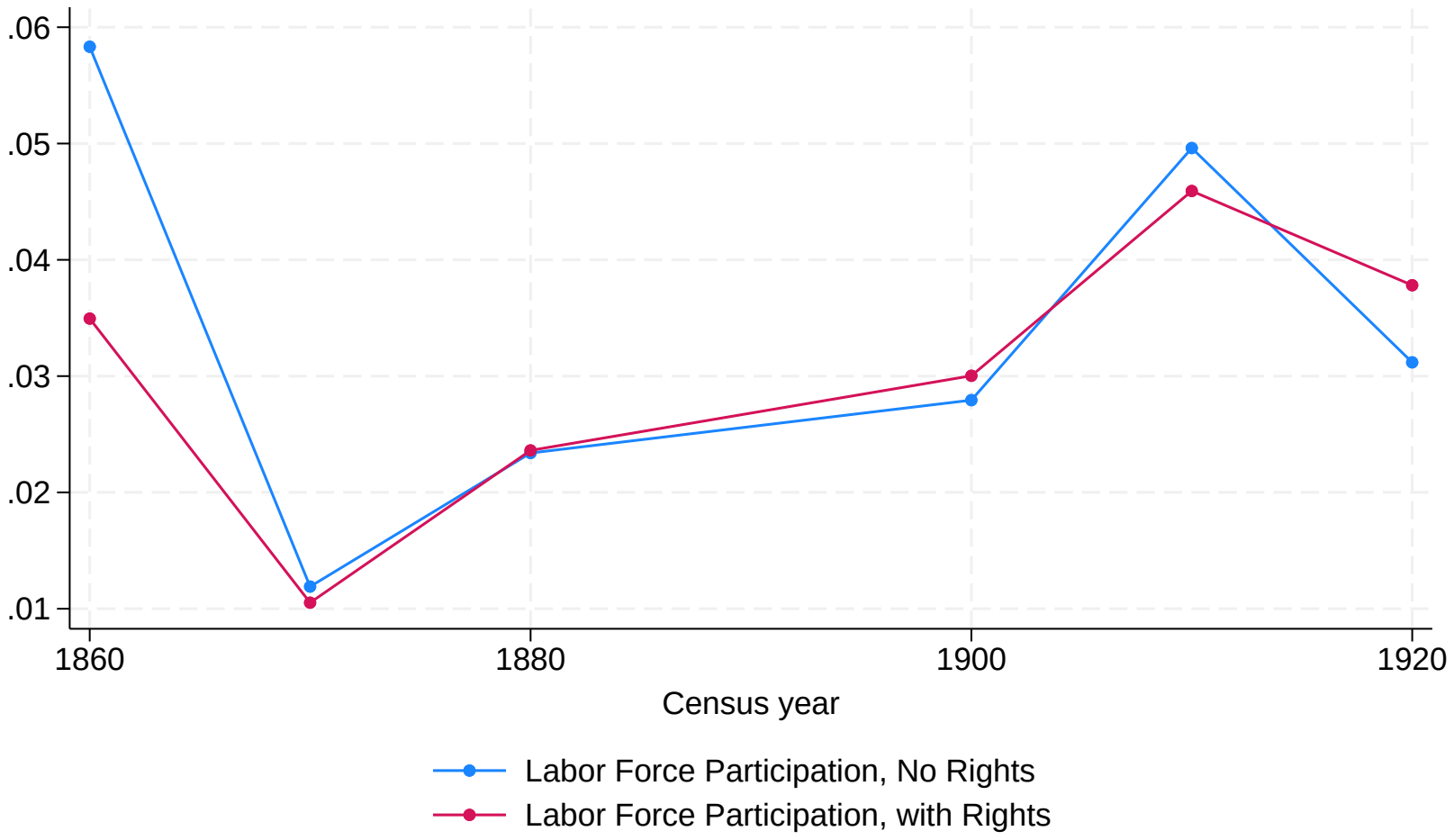
Birth Last Year & # of Kids Under Age 5, by Wealth 1860-1870

Dependent Variable	Birth Last Year		# Children Under Age 5	
	(1)	(2)	(3)	(4)
Rights	-0.007***	-0.006**	-0.016**	-0.005
	(0.002)	(0.003)	(0.007)	(0.009)
High Wealth		-0.003		-0.008
		(0.002)		(0.011)
High Wealth × Rights		-0.006*		-0.047**
		(0.004)		(0.018)
Controls	Yes	Yes	Yes	Yes
Extra Controls	Yes	Yes	Yes	Yes
N	3,302,806	3,302,806	3,302,806	3,302,806
Adjusted R^2	0.020	0.020	0.106	0.107
Mean Dep. Var.	0.224	0.224	1.313	1.313

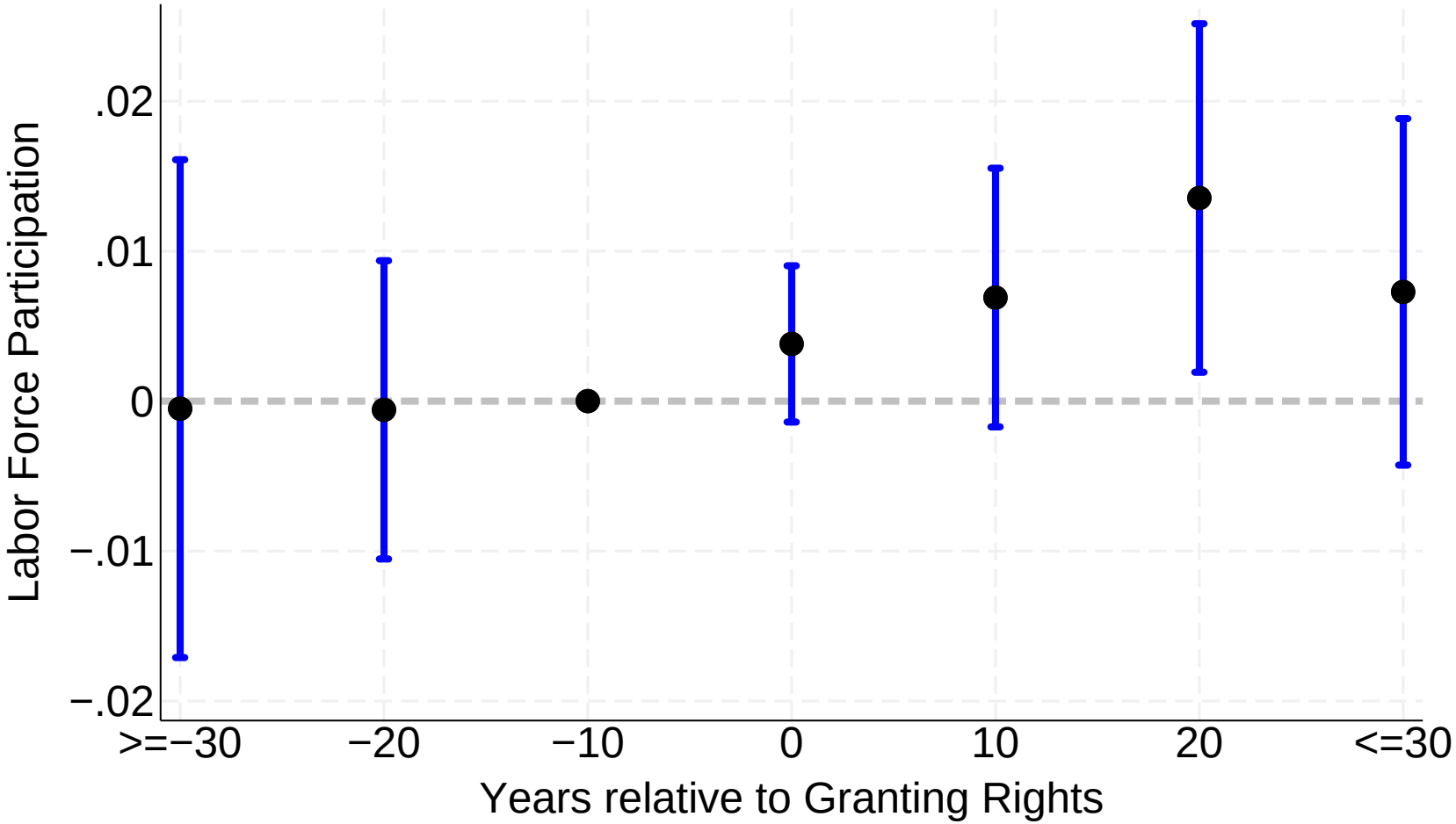
Labor Force Participation Ages 20-39, 1860-1920



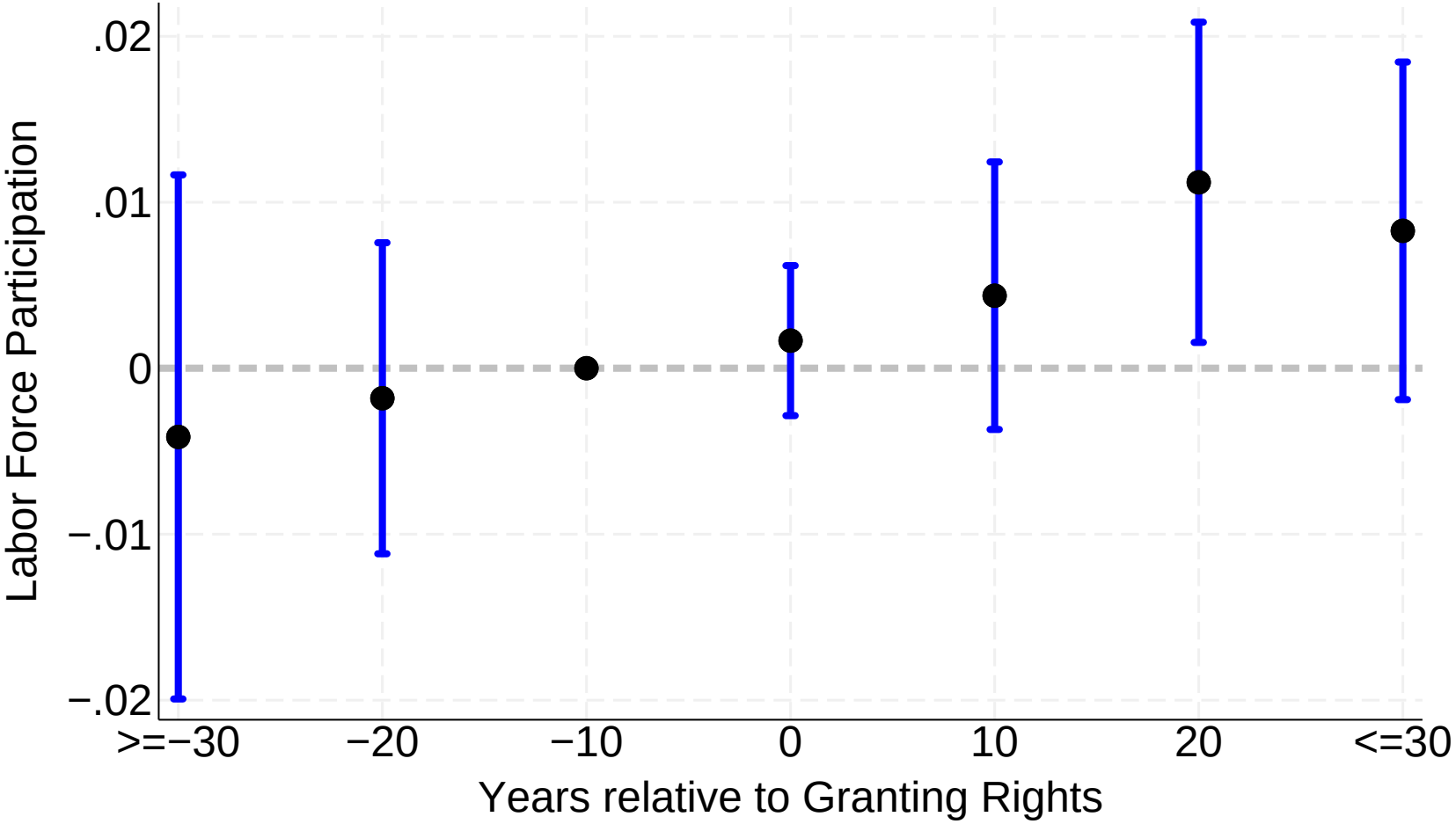
Labor Force Participation Ages 40-59, 1860-1920



Labor Force Participation Ages 20-39, 1860-1920



Labor Force Participation Ages 40-59, 1860-1920



Labor Force Participation, Married After Rights 1900-1910

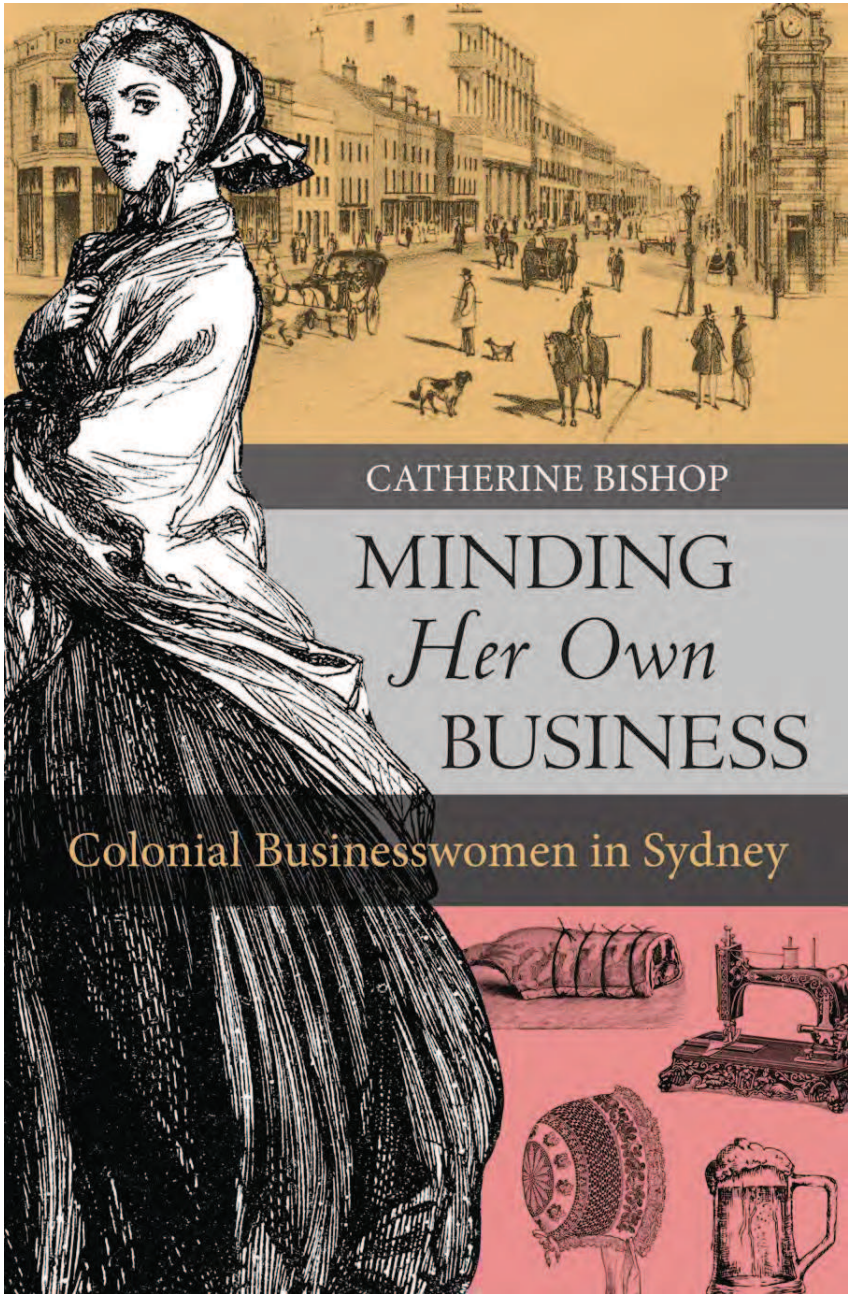
	(1)	(2)	(3)	(4)
Dep Var:	Labor Force Participation			
	Women Age 20-39		Women Age 40-59	
Married After Rights	-0.000	0.001	0.000	0.000
	(0.001)	(0.001)	(0.001)	(0.001)
Controls	Yes	Yes	Yes	Yes
Extra Controls	Yes	Yes	Yes	Yes
Sample	All	Stayers	All	Stayers
N	10,040,159	7,258,587	3,485,362	2,266,292
Adj. R^2	0.053	0.063	0.035	0.050
Mean Dep. Var.	0.034	0.033	0.036	0.043

Concluding Remarks

- We found that women's rights had a dramatic impact on the family:
  - Less fertility
  - No change in LFP
  - Mechanism: household bargaining.
- On going work: Rights led to about a 7-10% increase in schooling of children.
  - Equal for sons/daughters; Mostly for kids ≤ 13
  - About 20% of increase in this time period (1850-1920)
  - By linking to their self adult via Census Tree, increase in completed education and income in 1940.
  - Next, look at 2nd generation fertility and 3rd generation education ...

Thank you!

Coverture in New South Wales



Coverture in New South Wales

- Wives of free men were regarded under common law as femmes covert – Similar to England and the US.
- Wives of attainted men were regarded as femmes sole; temporary: at the end of a man's sentence, his wife reverted to being a femme covert.
- Some issues with wives of free men; socially higher in status; Odd to deny them from rights of wives of convicted men (Bishop 2015).

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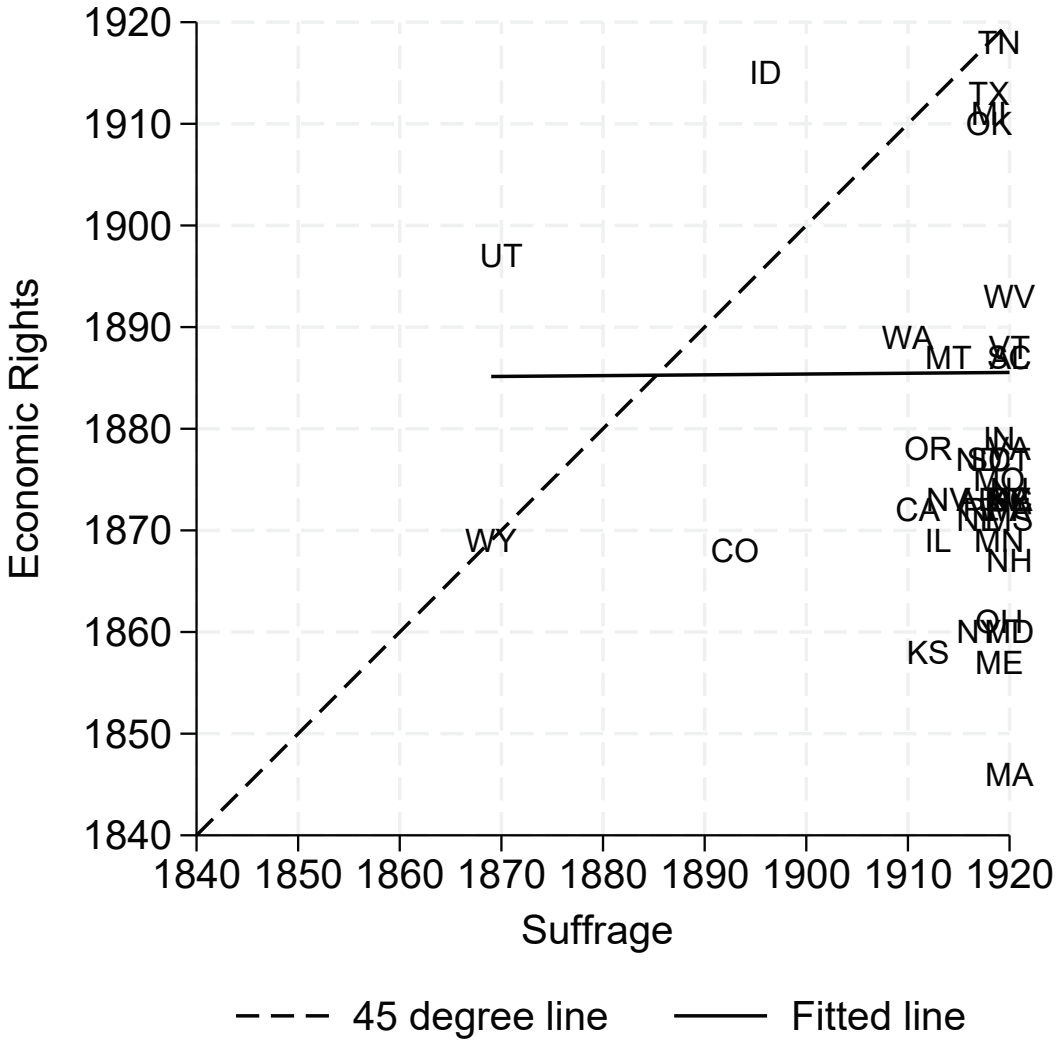
Timing of Women's Rights by State/Type



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Community Property

- 8 states had "community property": AZ, CA, ID, LA, NV, NM, TX, WA.
- Based on Spanish civil law (LA on French civil law).
- 3 types of property: Wife, husband, community.
- Wife had 50% interest in community, 100% in her separate property. No control.
- In principle, different. In practice? (Schuele 1994)
  - Immigrants didn't understand common law not in effect.
  - Even lawyers/lawmakers didn't understand.
  - Men often used women's property for their own benefit.
- Regardless, women's rights had impact.

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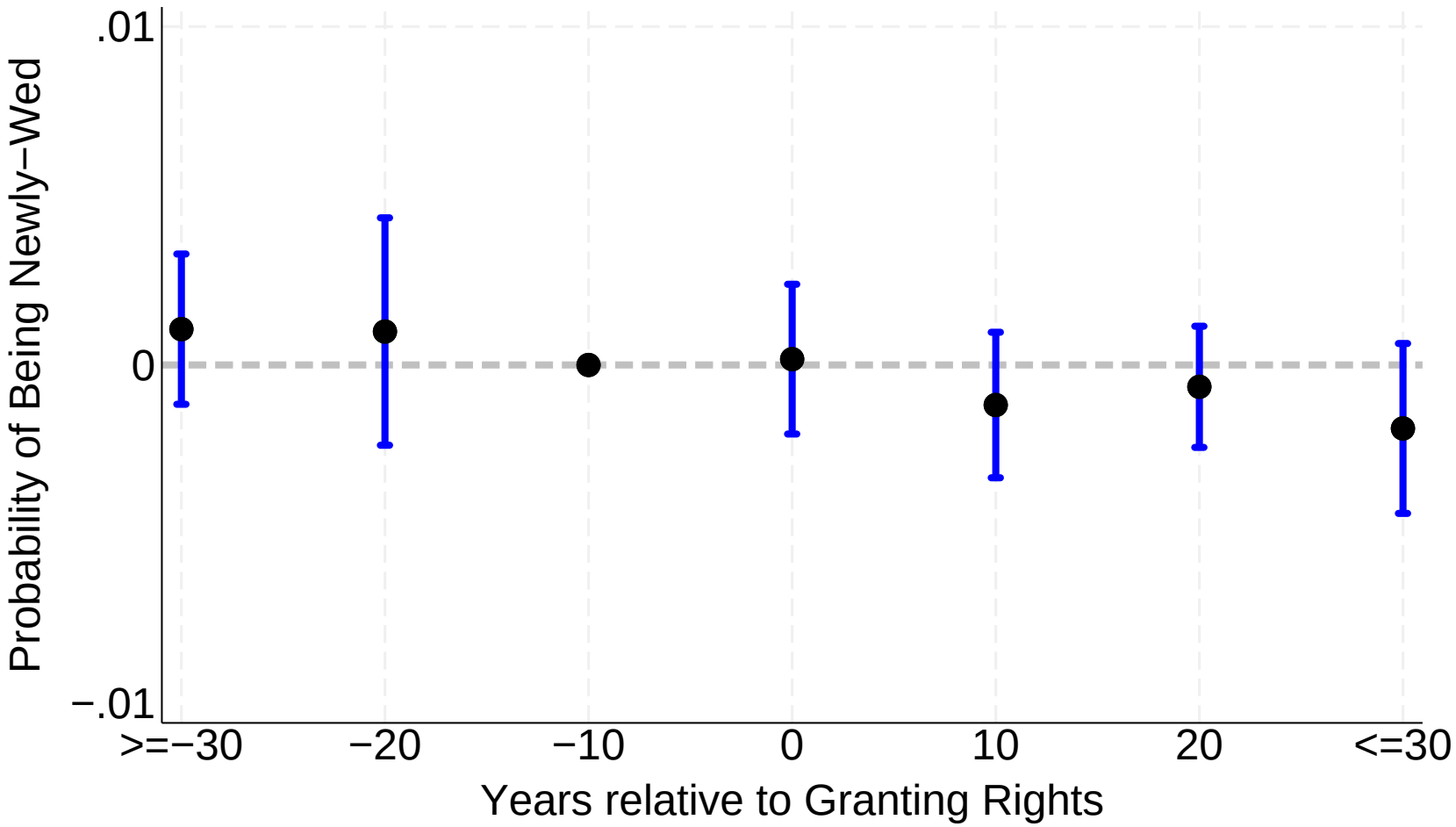
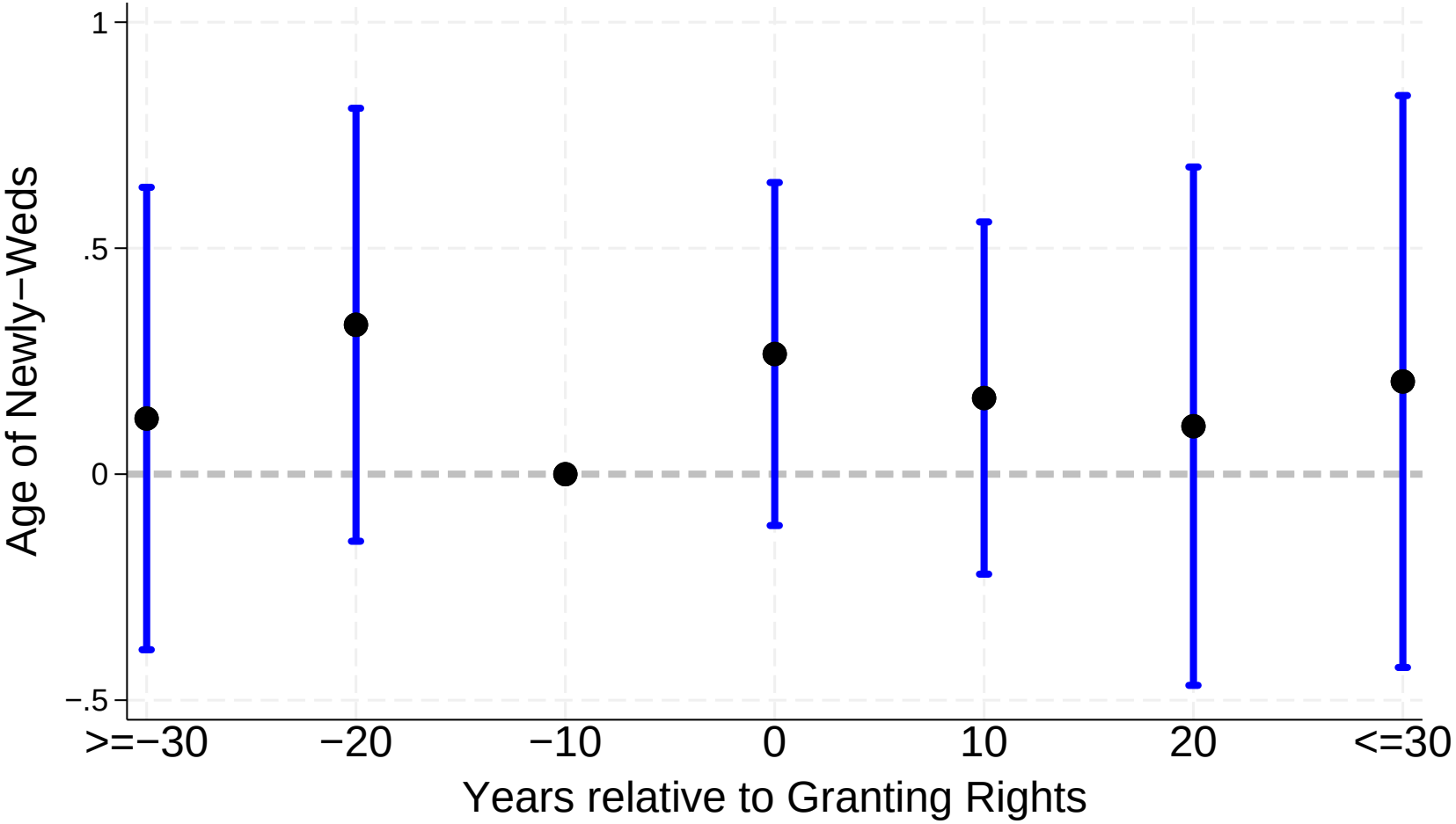
The Marriage Market for Newly Weds: 1850-1910

	Marriage Market-Men 20-50		
Dependent Variable	Prob NW	Age NW	Age-Gap NW
	(1)	(2)	(3)
$\geq$ 3 Decades Before	0.001	0.123	-0.185
	(0.001)	(0.261)	(0.131)
2 Decades Before	0.001	0.321	-0.024
	(0.002)	(0.244)	(0.090)
1 Decade Before	0	0	0
Rights Given	0.000	0.266	-0.039
	(0.001)	(0.194)	(0.130)
1 Decade After	-0.001	0.168	0.158
	(0.001)	(0.199)	(0.201)
2 Decades After	-0.001	0.107	- 0.014
	(0.001)	(0.293)	(0.248)
$\geq$ 3 Decades After	-0.002	0.205	-0.239
	(0.001)	(0.323)	(0.292)
N	20,107,591	577,366	577,366
Adj. R^2	0.069	0.101	0.038
Mean Dep. Var.	0.029	27.54	4.39

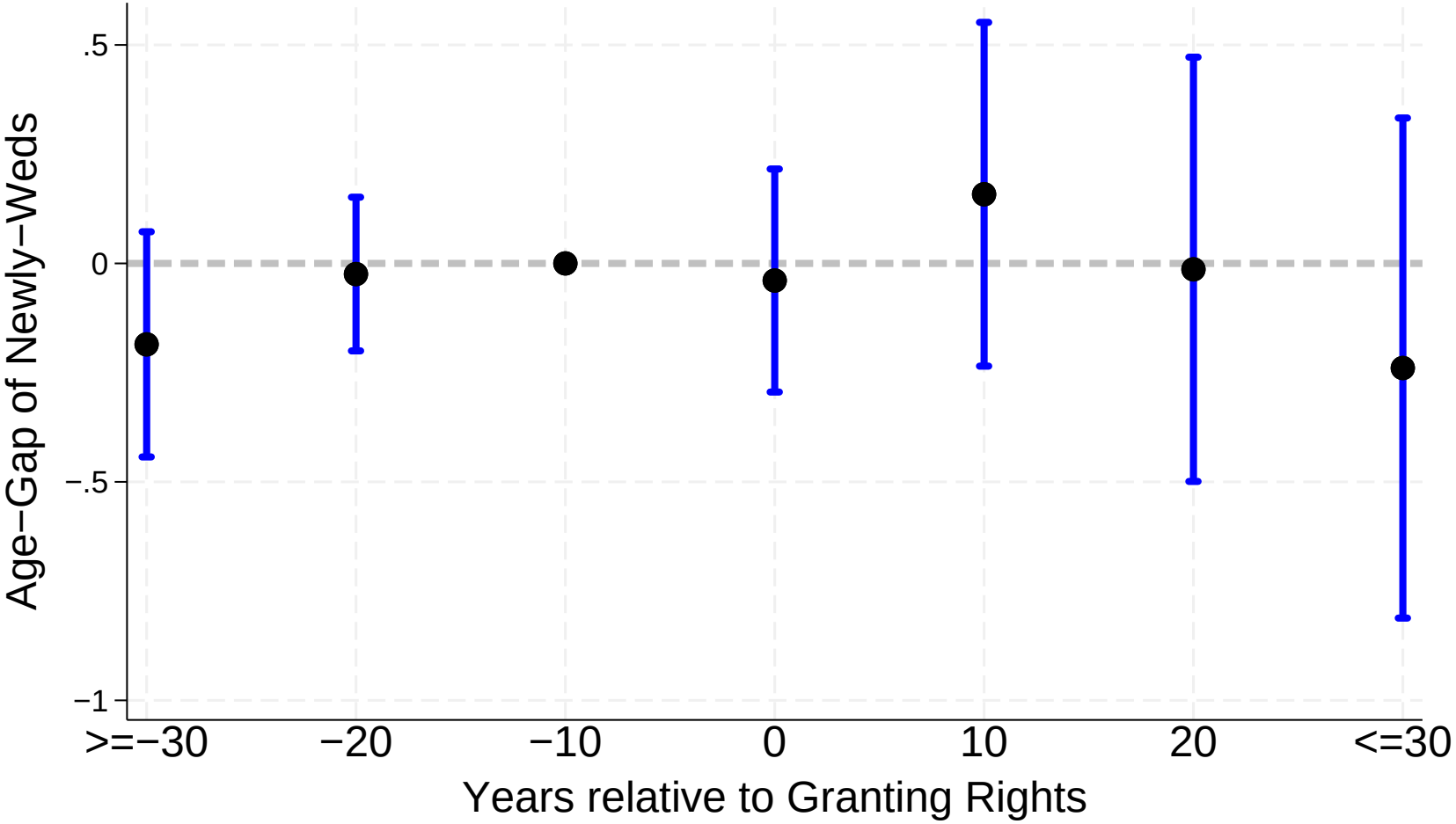
Probability of being Newly Wed: Men 20-50, 1850-1910



Age of Newly Wed: Men 20-50, 1850-1910



Age Gap btwn Newly Wed Couples: Men 20-50, 1850-1910



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